L1-Basics of Programming

Special class

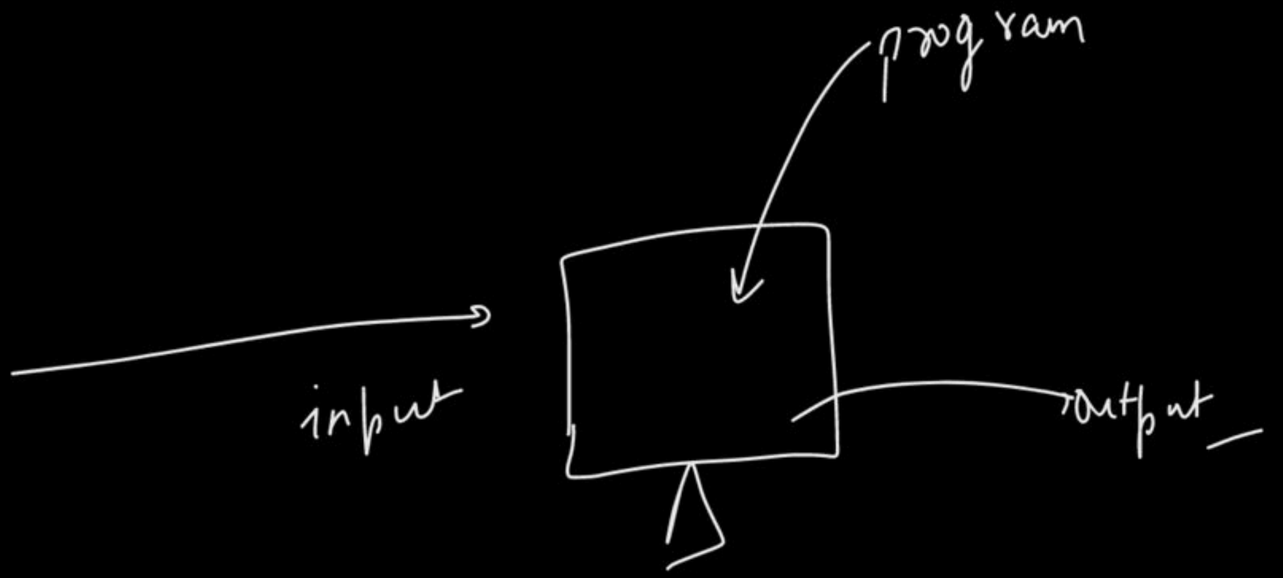
Love Babbar • Apr 2, 2025

L1 - Basics of Programming

Flowcharts and PseudoCode

—> by Codehelp

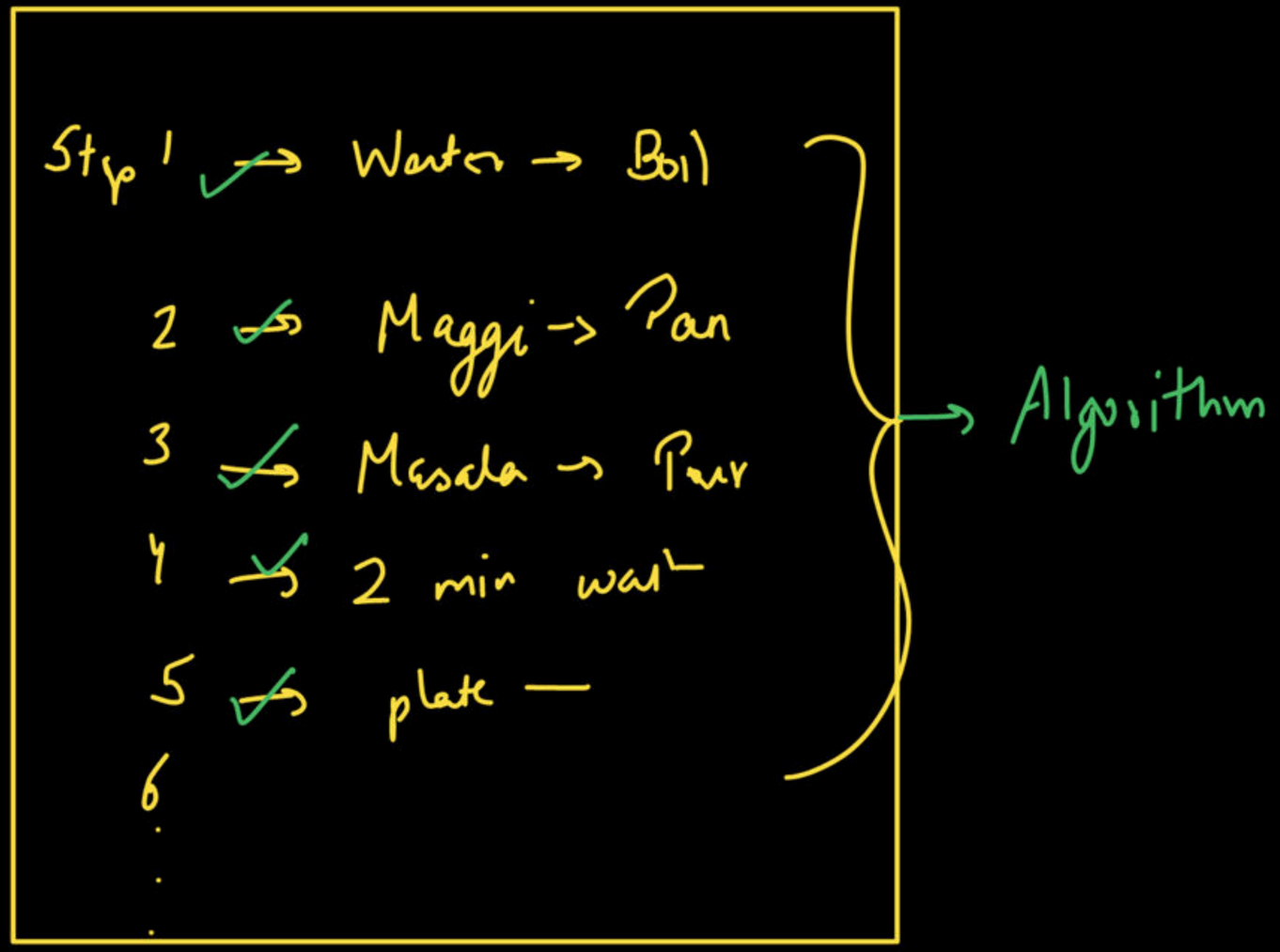
What is Programming ?

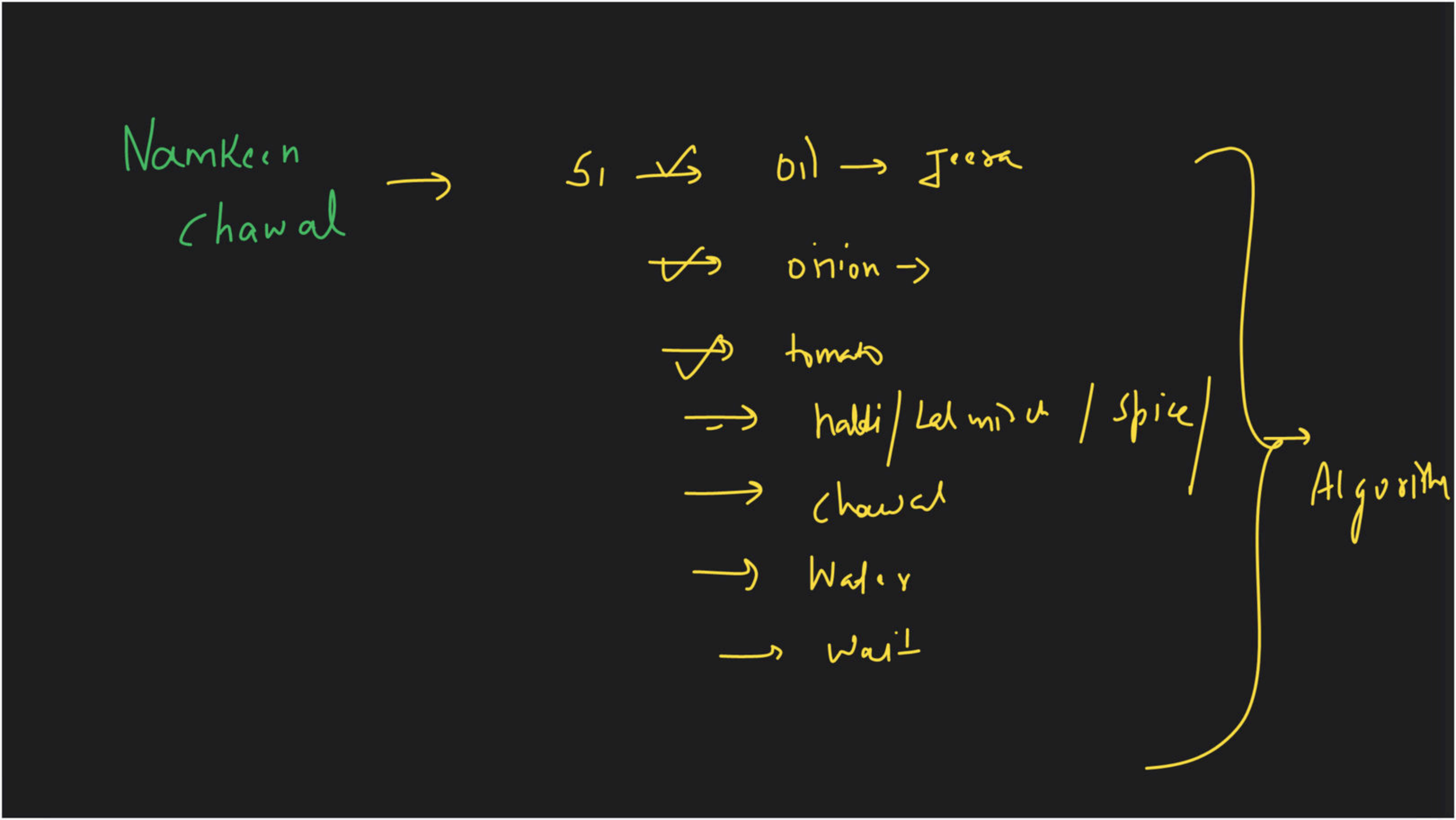


What is Algorithm ?

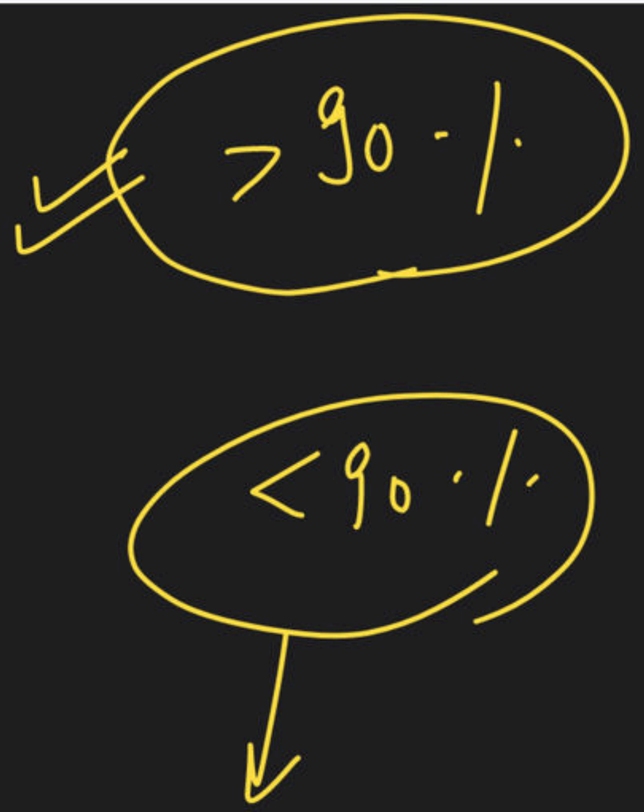
Maggi →

Shrey Vohra




Namkeen Chawal
Oil → Jeera
onion →
tomato
haldi / Lal mirch / spice /
chawal
Water
Salt
Algorithm

Pasta → ?



How to approach a Problem ?

Thought process

① Lets understand the Problem

② Analyse problem → given values / relate formula / constraints

side = 4 cm

③ Create a approach → Algorithm

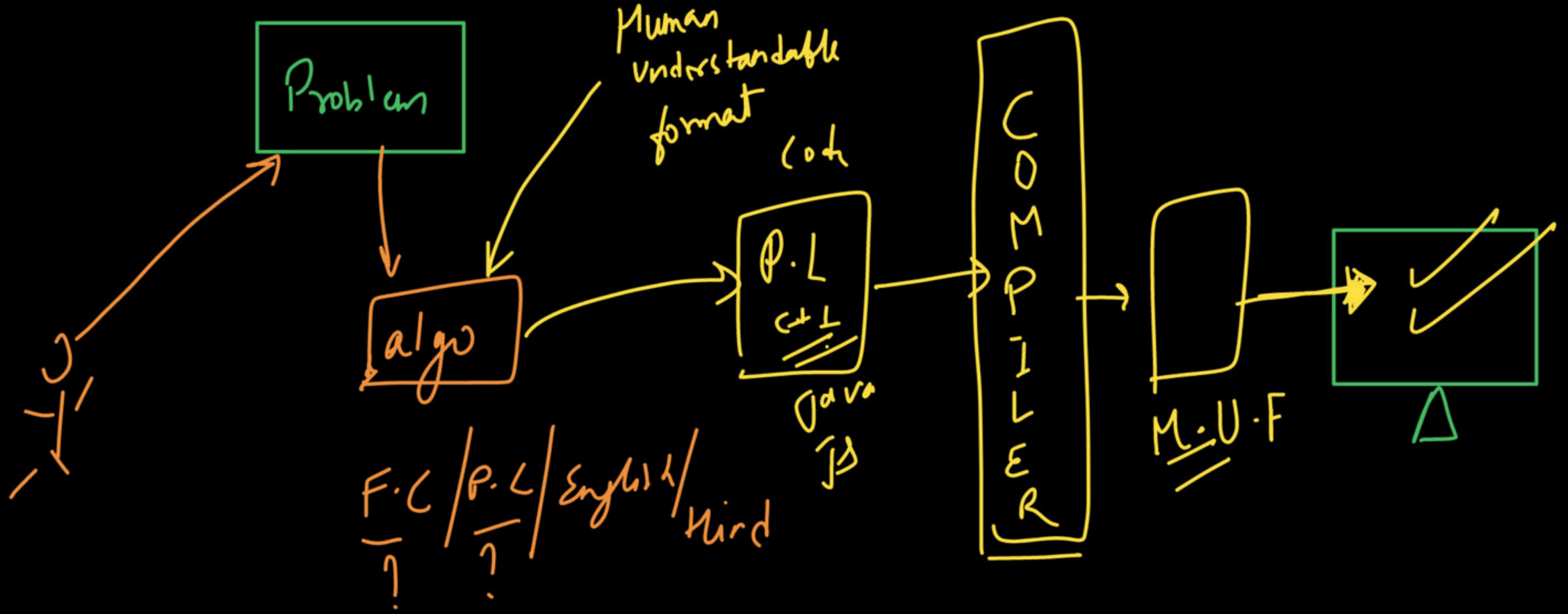
Optimise

area = s × s

= 4 × 4

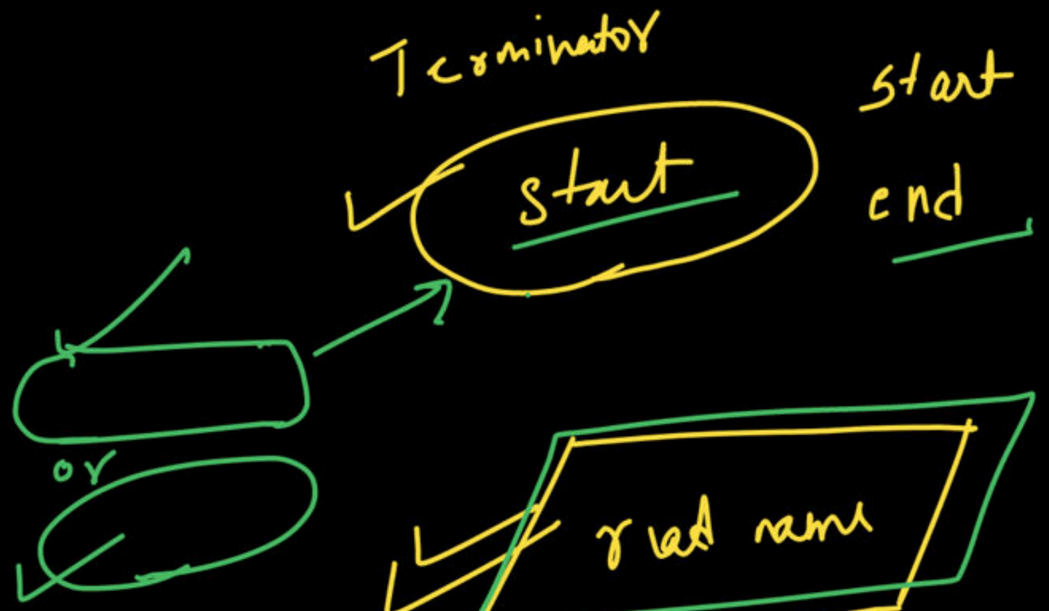
= 16

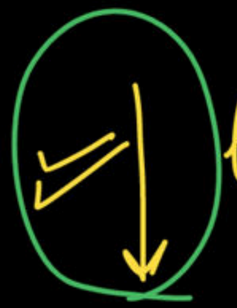
Using a computer to solve a problem



Flowchart and its components:

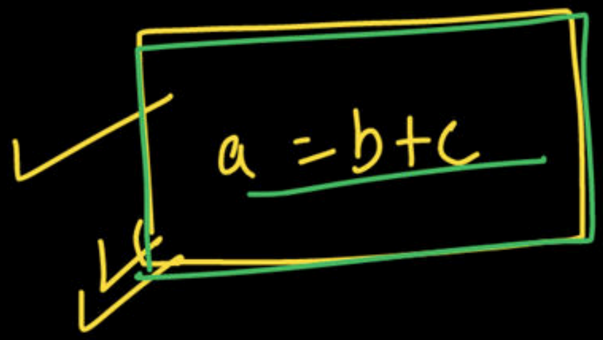
Diagrammatic representation of your Algorithm



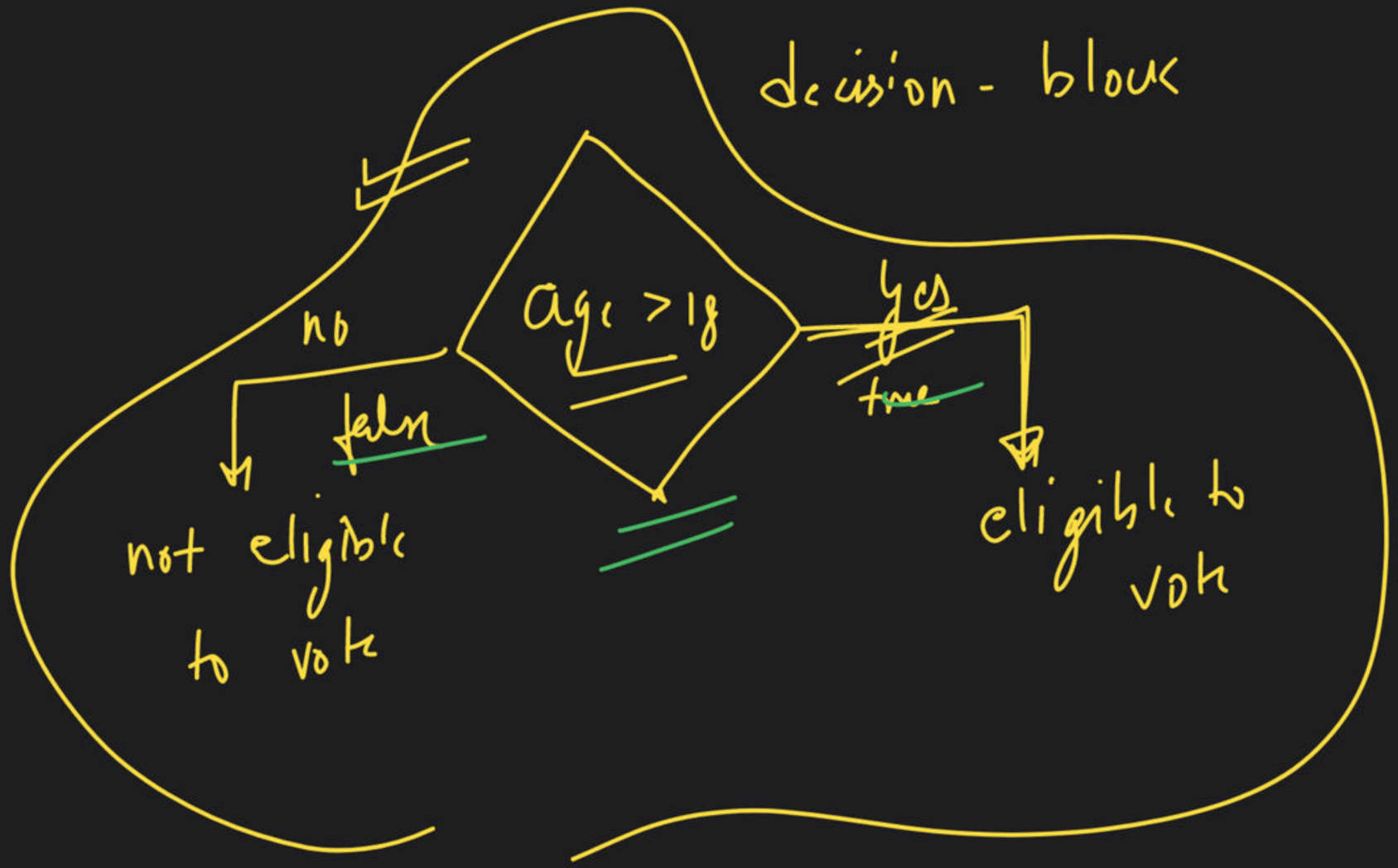


flow of execution

read
input / print
Output



→ process block
processing / operation / calculation


decision - block
age > 18
no
false
not eligible to vote
yes
true
eligible to vote

Pseudocode:

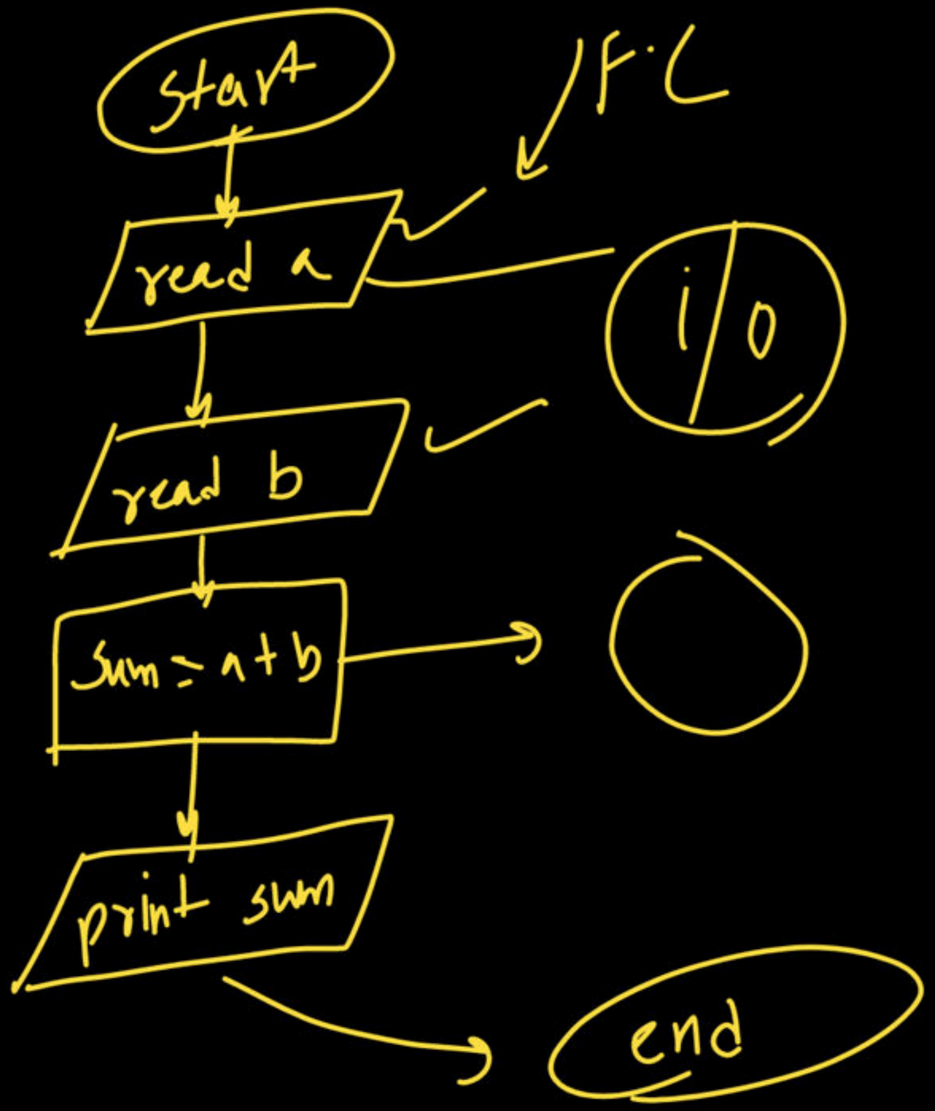
Nakli → code → denote → dui bhasha

Sum of 2 no

- start
- read the value of a & b
- sum = a + b
- print sum
- end

- start
- read first name
- read middle name
- read last name
- f.n = f + m + l
- print full name
- end

Design Flowchart - Print Sum of a and b



P.C

a = 10

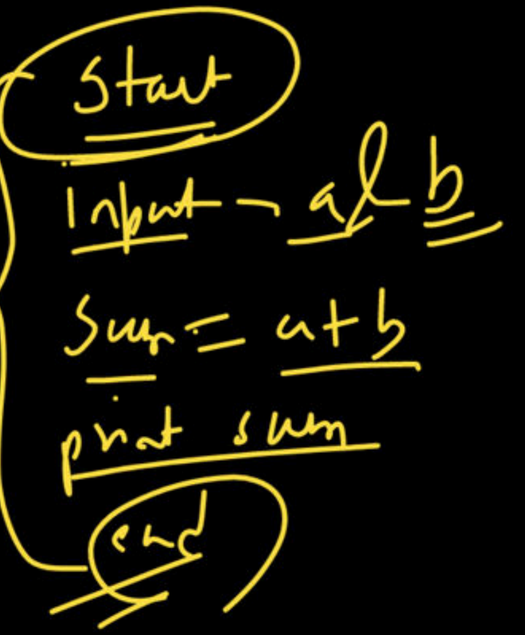
b = 5

Sum = a + b

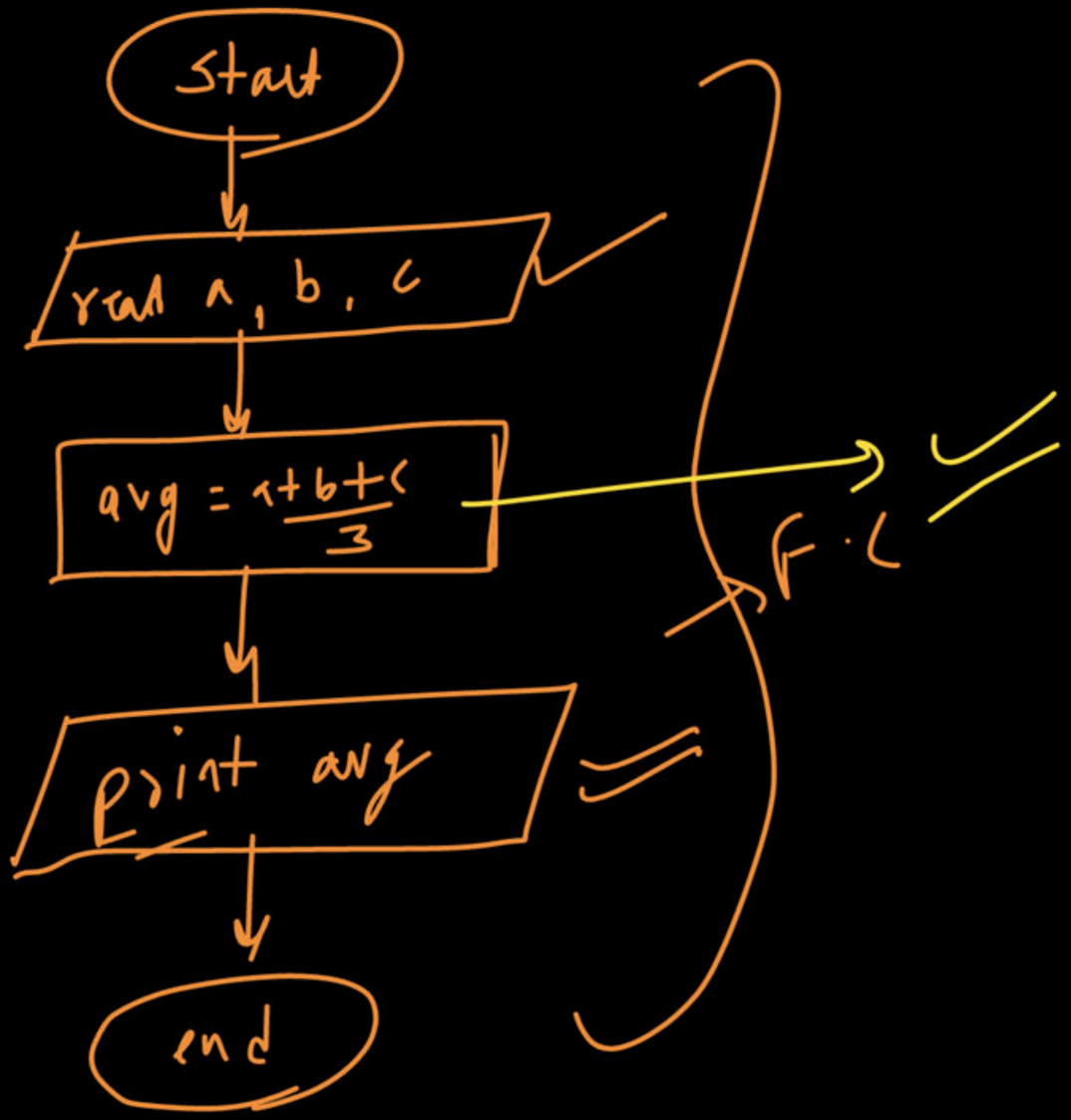
= 10 + 5

Sum = 15

print 15



Design Flowchart - Print Average of a, b and c



Start

read a = 10, b = 12, c = 14

avg = (a+b+c) / 3

= (10+12+14) / 3 = 36 / 3 = 12

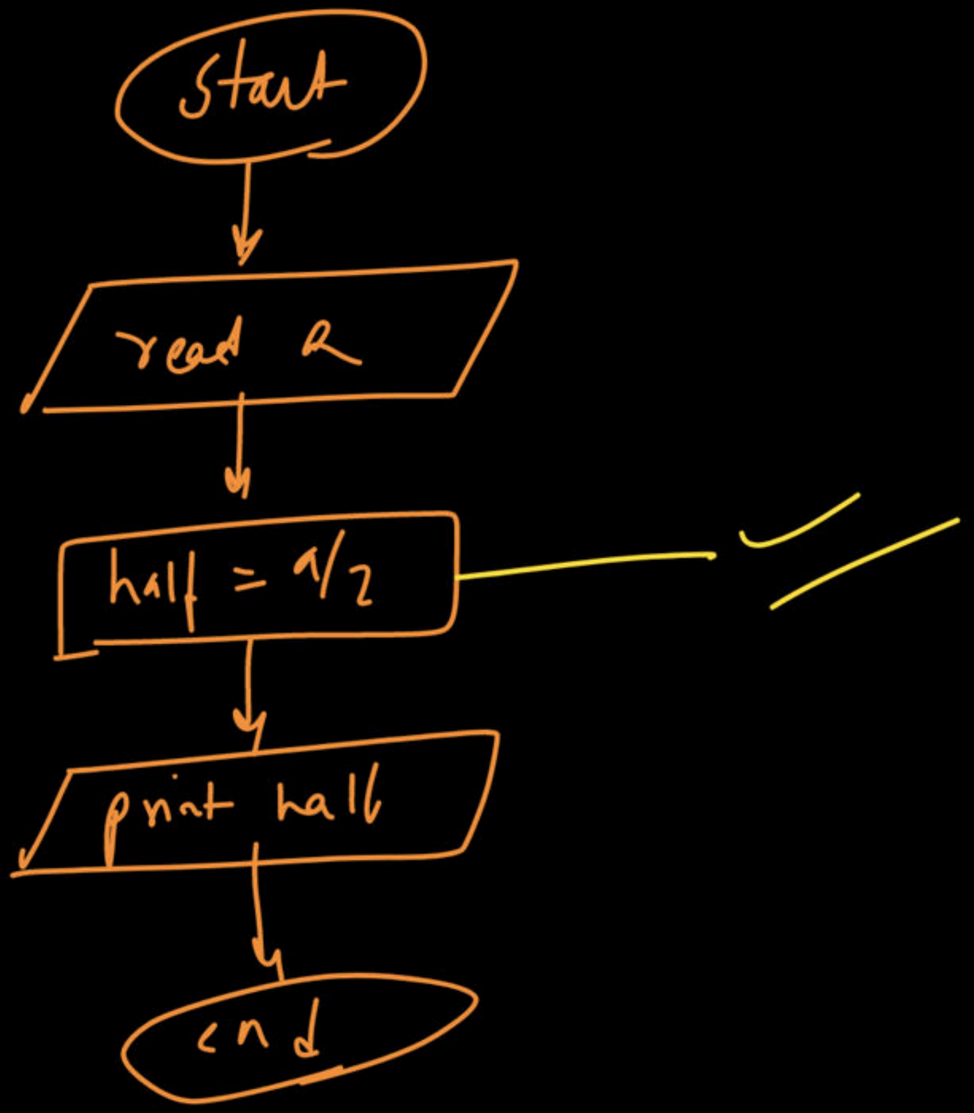
avg = 12

print avg

end

Arbit Pathak

Design Flowchart - Print half of a



start

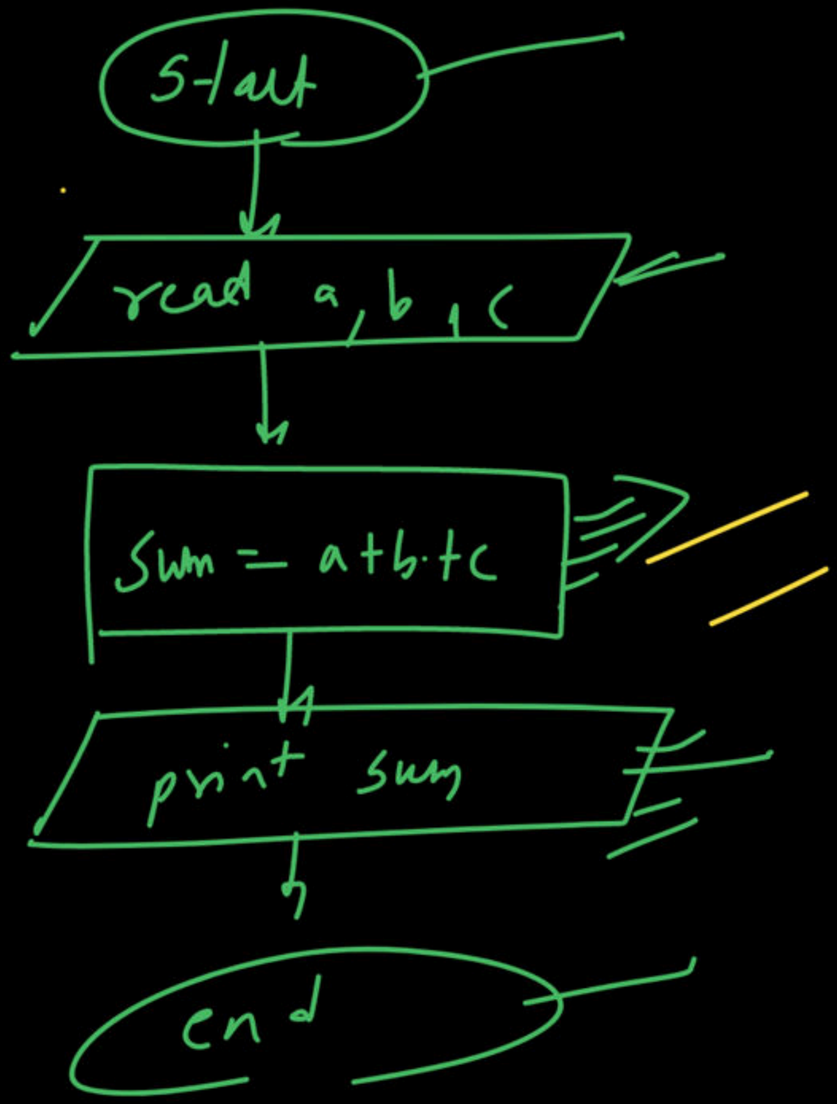
read a

half = a/2

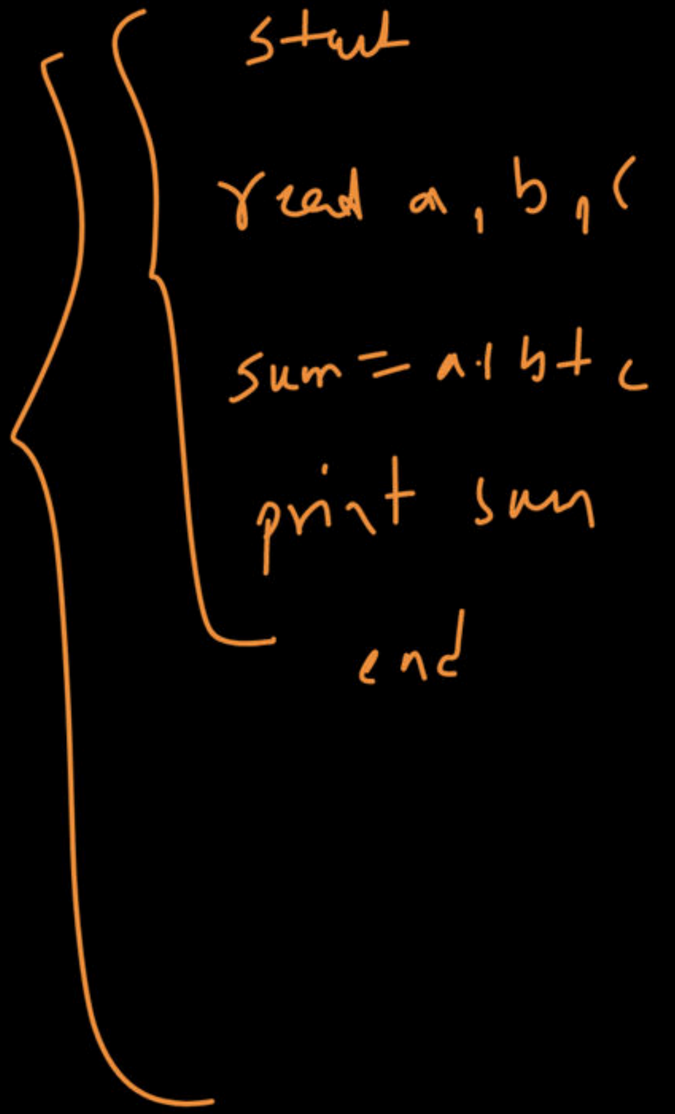
print half

end

Design Flowchart - Take Input and add 3 numbers

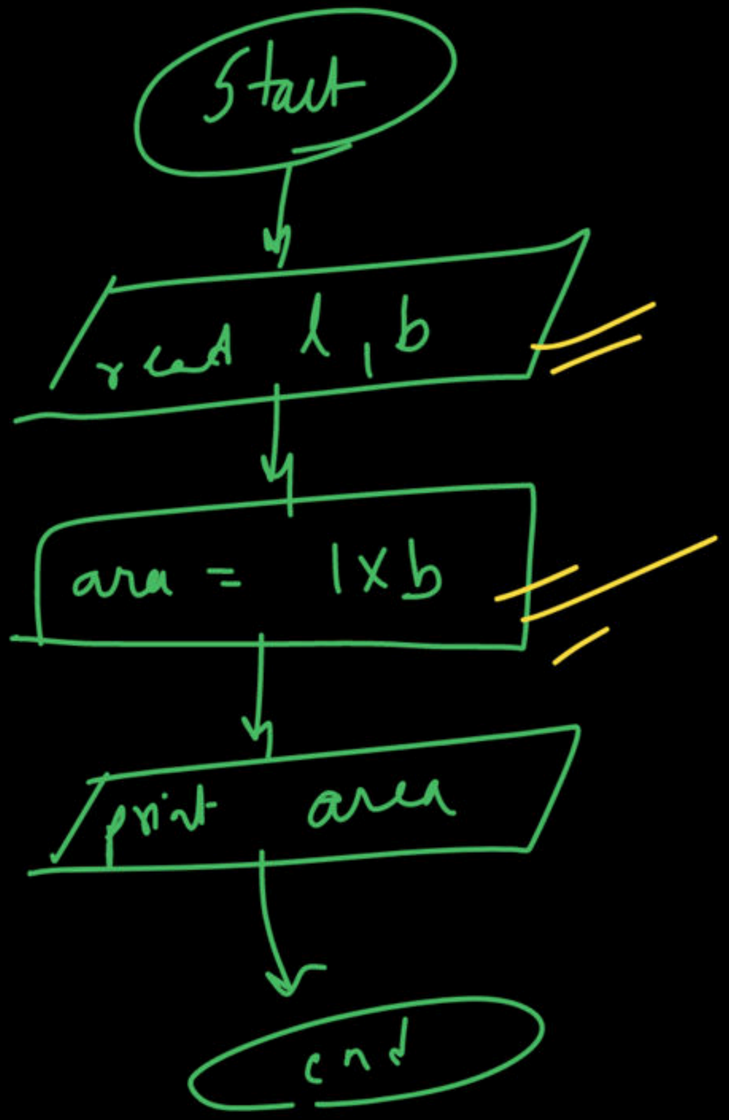






Design Flowchart - Area of Rectangle

area = len × br





Start

read len, br

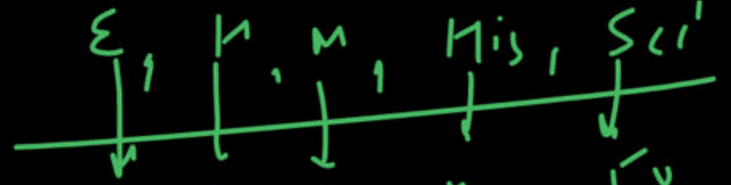
area = l x b

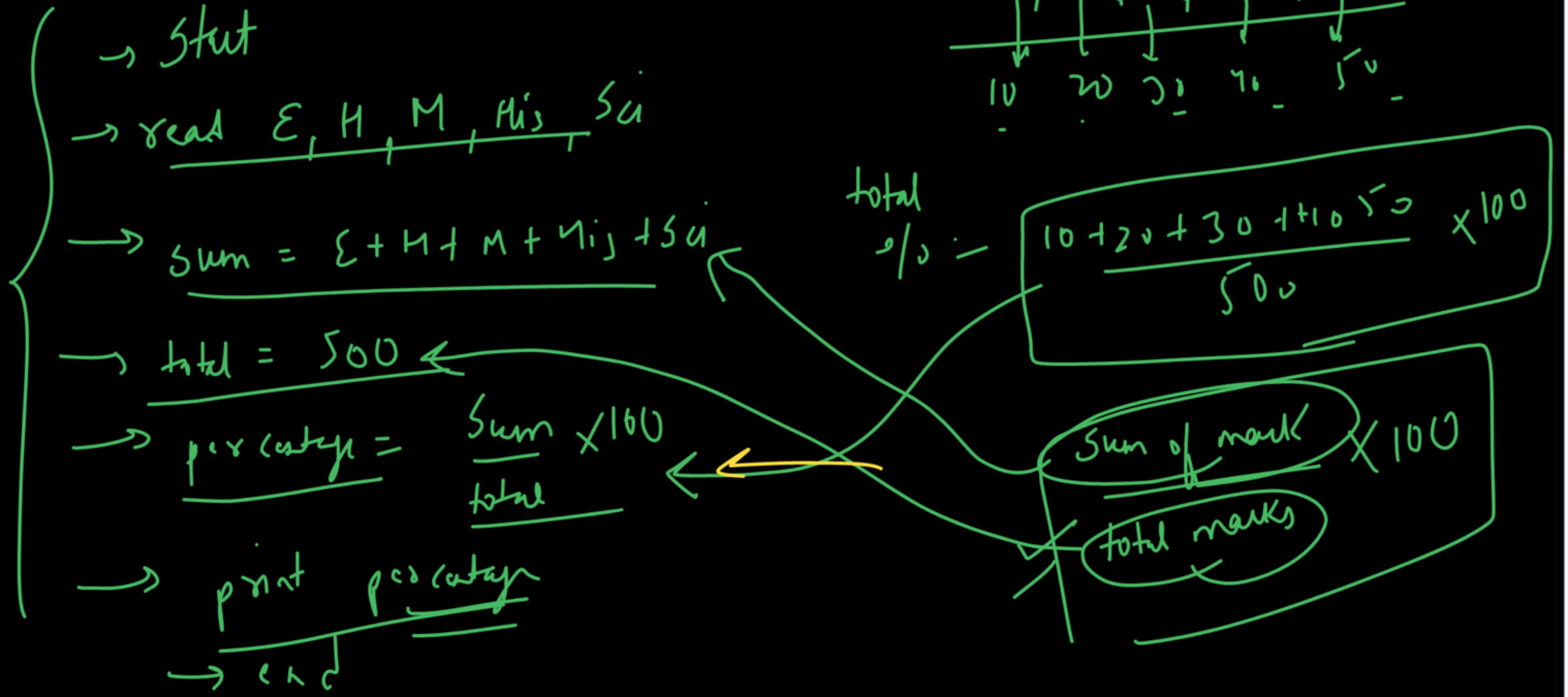
print area

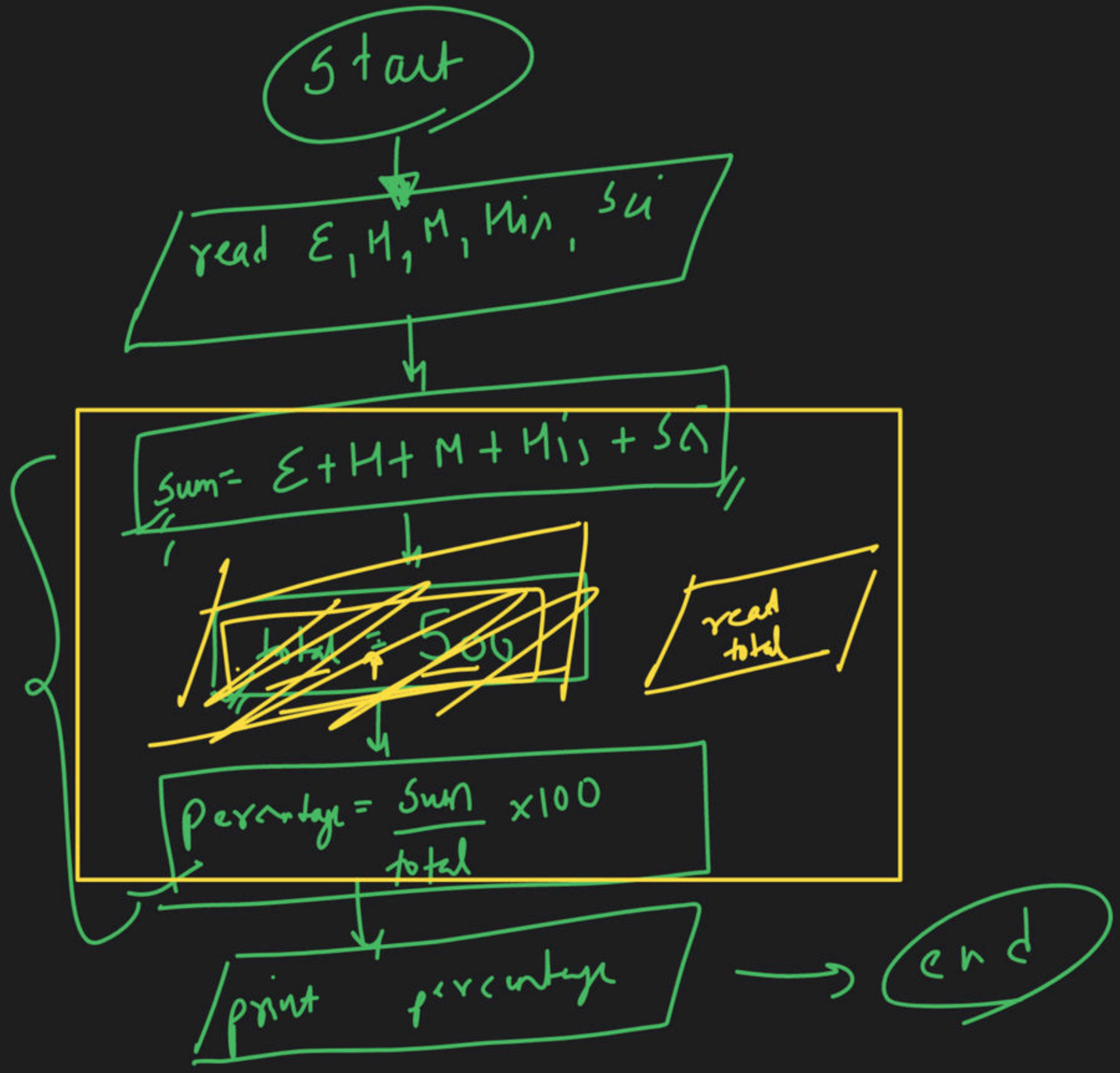
end

2 min

Design Flowchart - Calculate Percentage





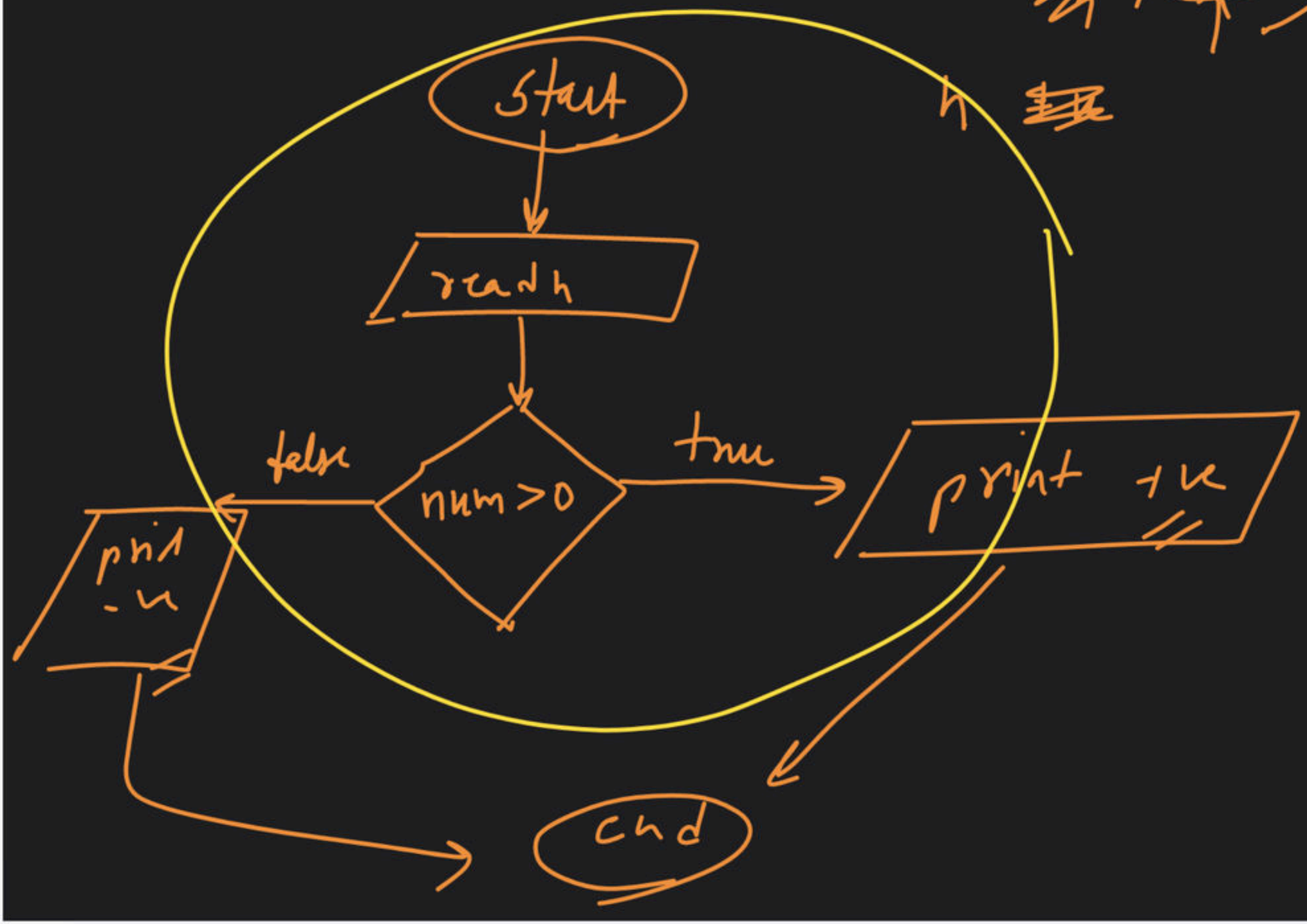

Start
read E, H, M, Mis, Sci
Sum = E + H + M + Mis + Sci
total = 500
read total
Percentage = Sum / total x100
print percentage
end

no → +ve / -ve

0 , -ve

+ve → >0

-ve → <0



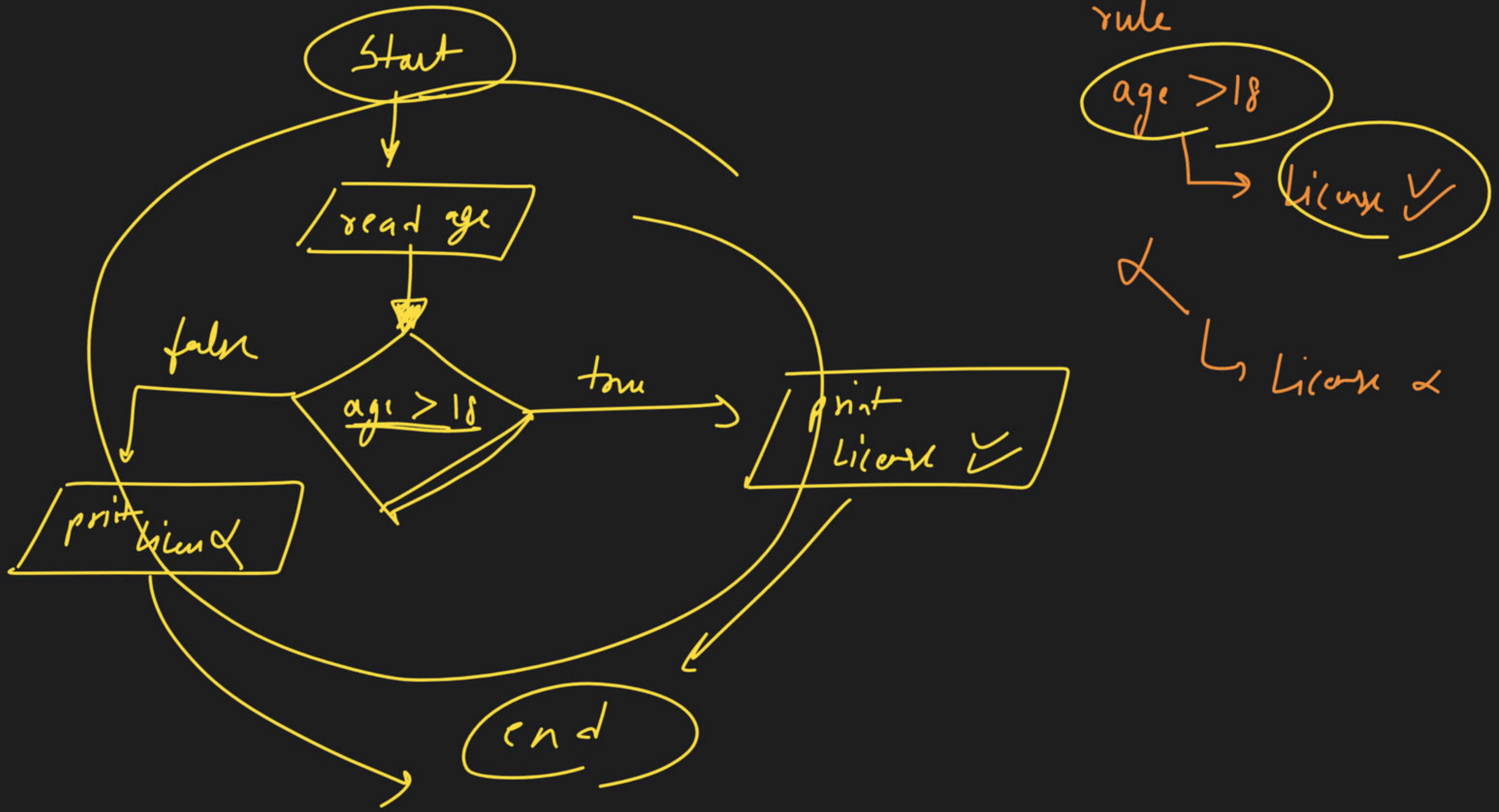
Start

read num

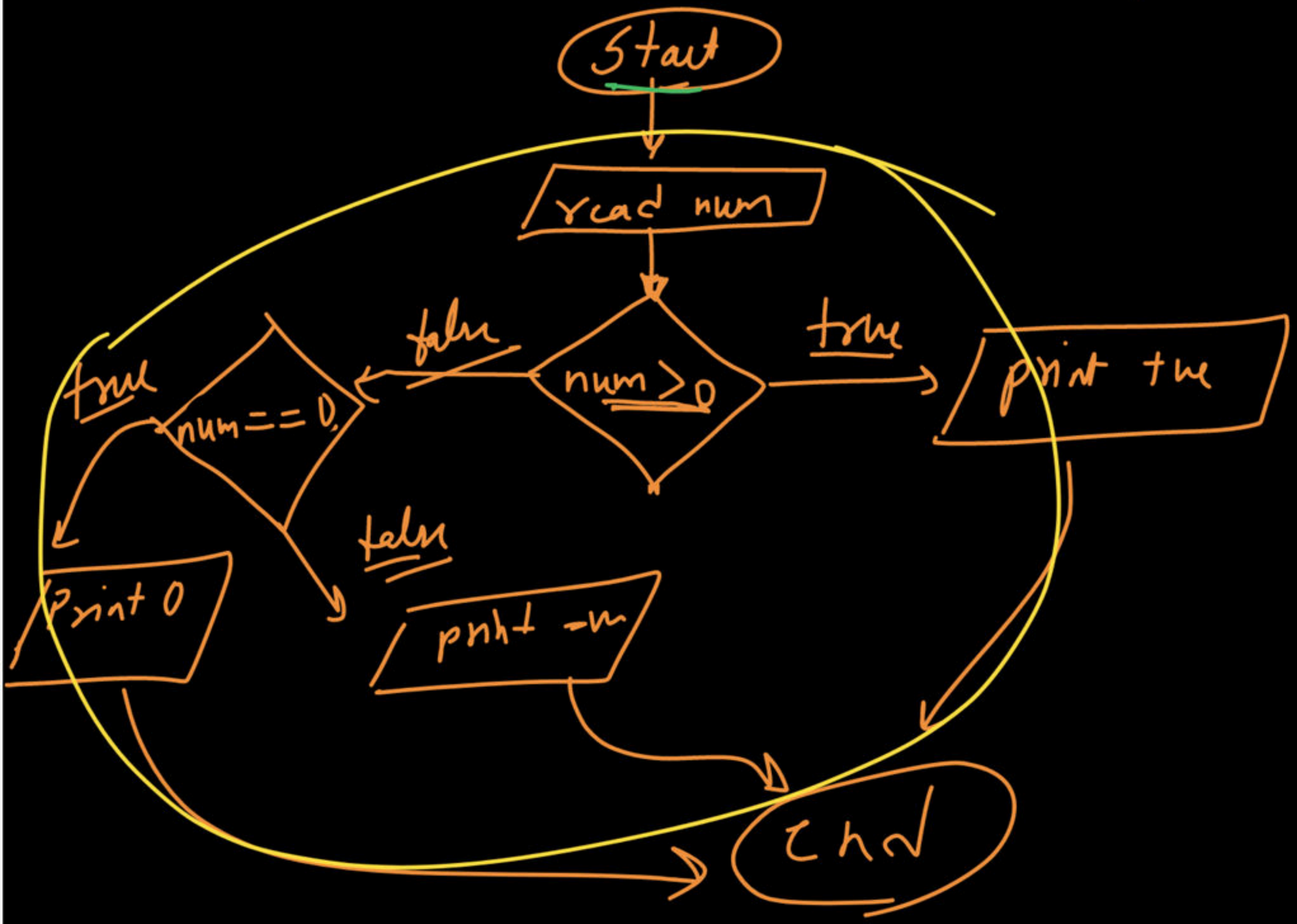
if num > 0, print +ve

if num < 0, print -ve

end


Start
read age
age > 18
false
true
print license ✗
print license ✓
end
rule
age > 18
license ✓
✗
license ✗

Design Flowchart - Check positive, negative or zero



Start

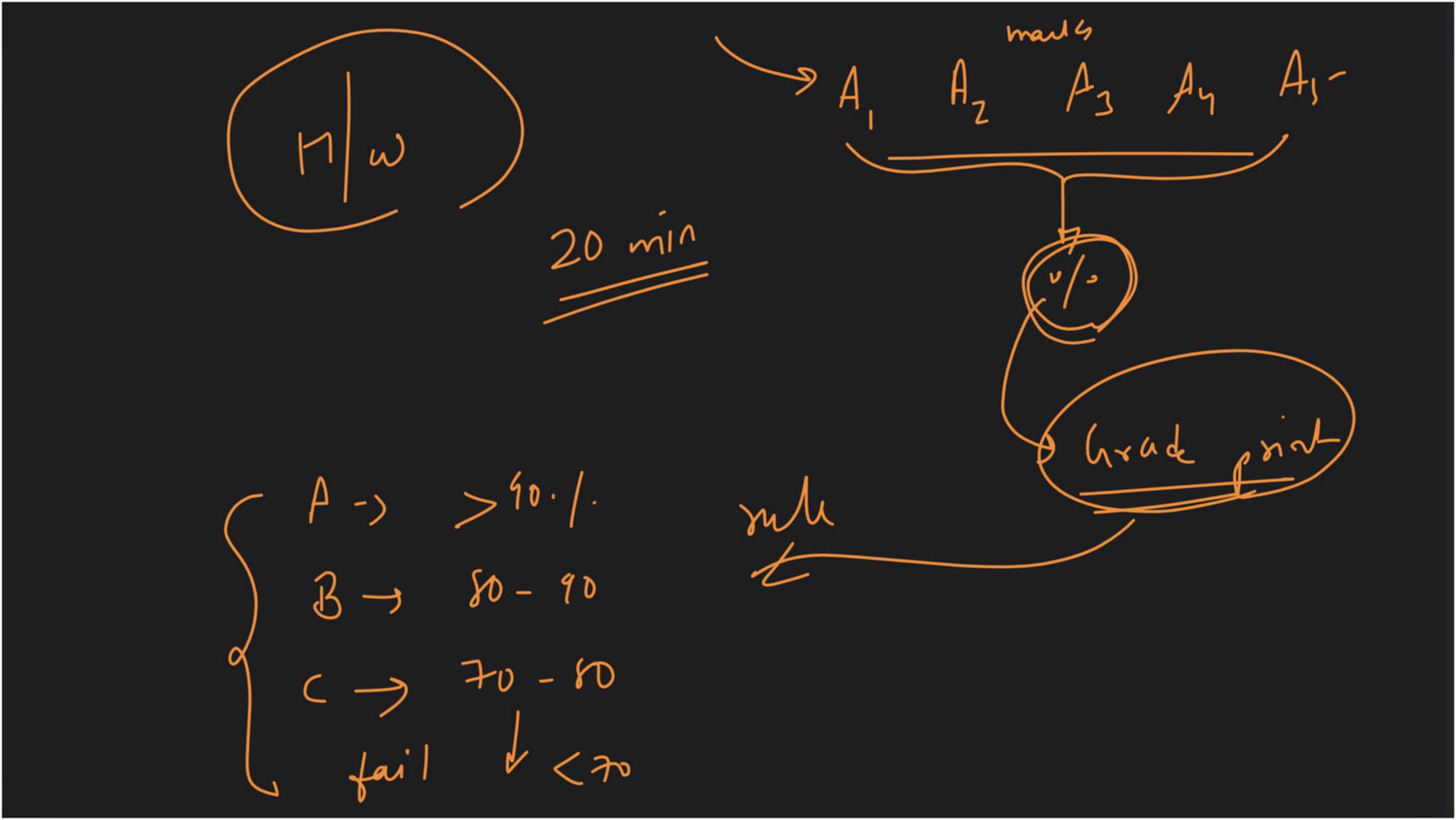
read num

if num > 0, print +ve

else if num = 0, print 0

else print -ve

end


H/W
marks
A1 A2 A3 A4 A5
20 min
V/C
Grade point
rule
A -> >90%
B -> 80 - 90
C -> 70 - 80
fail
<70

count = 0 1 2 3 4 5

limit = 5

Name ⇒ 5 print

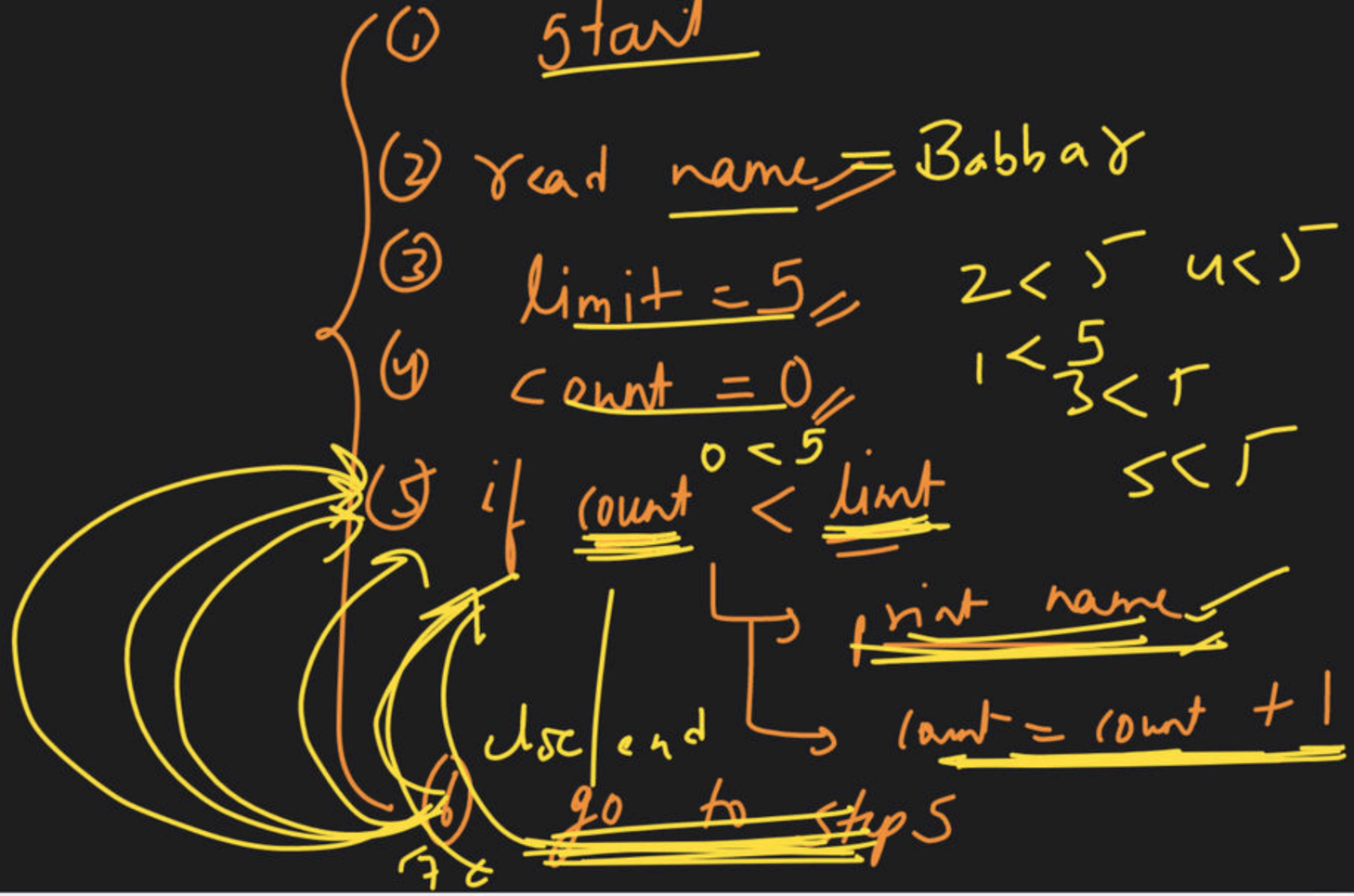
Babbar

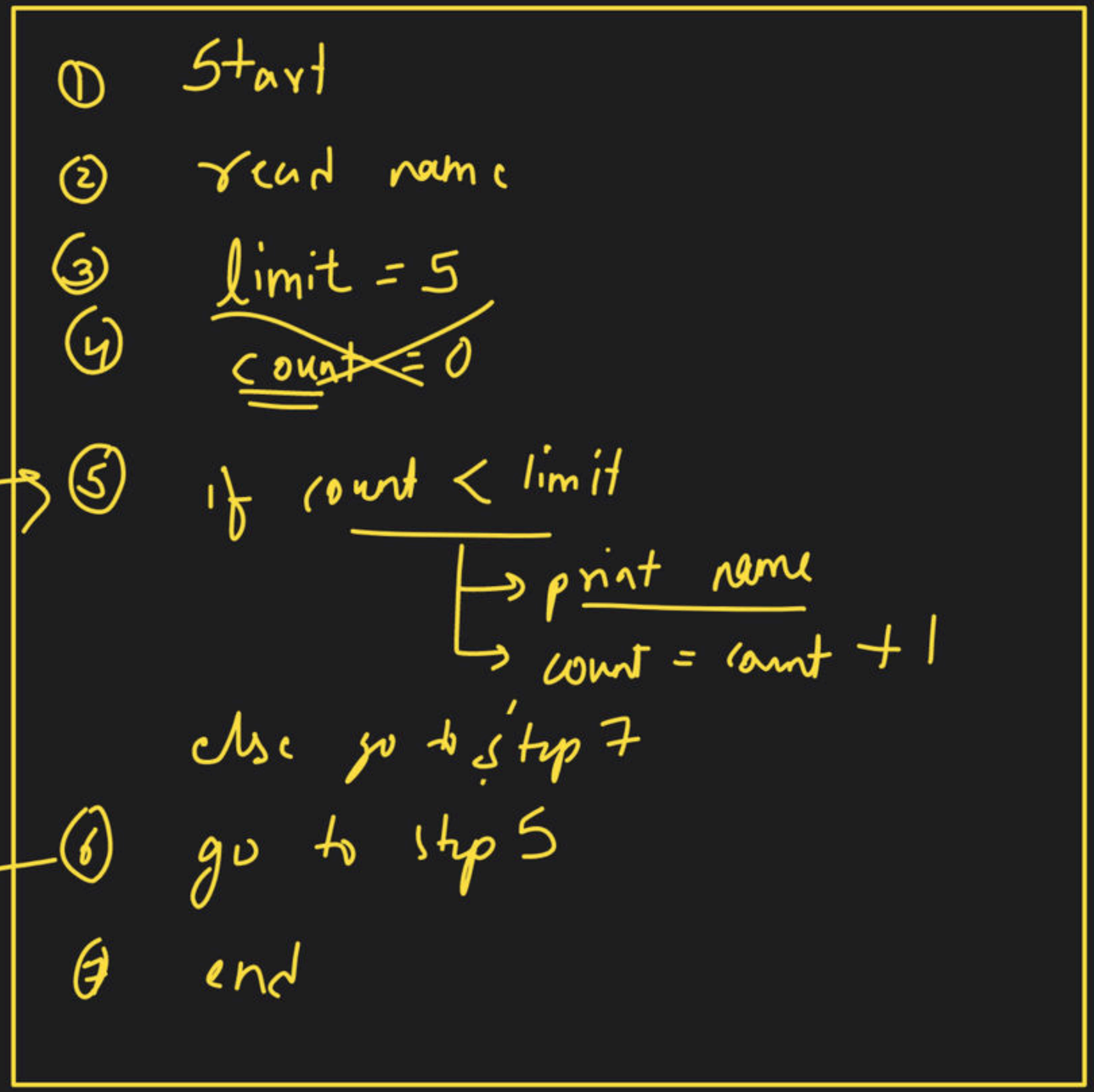
Babbar

Babbar

Babbar

Babbar

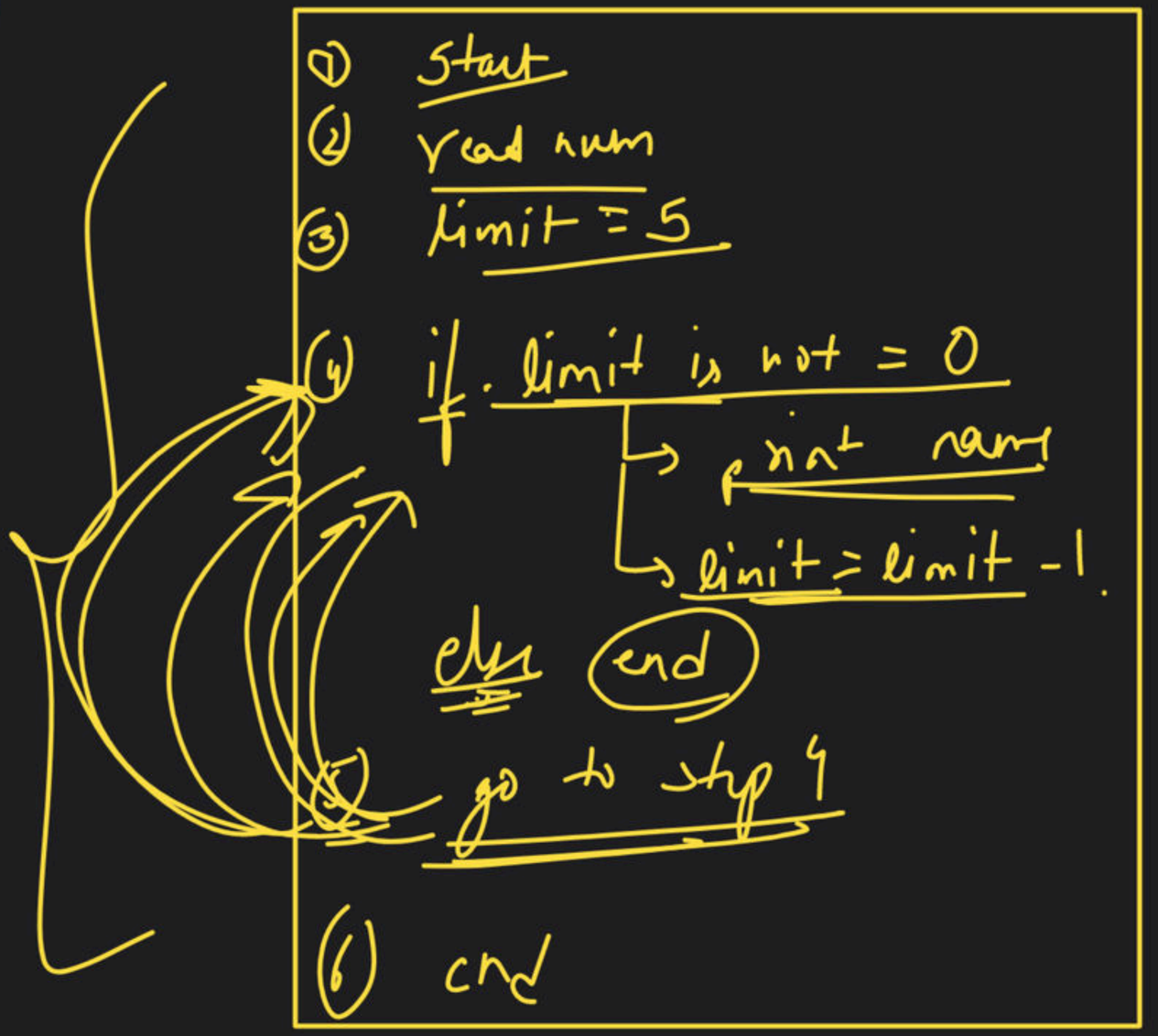


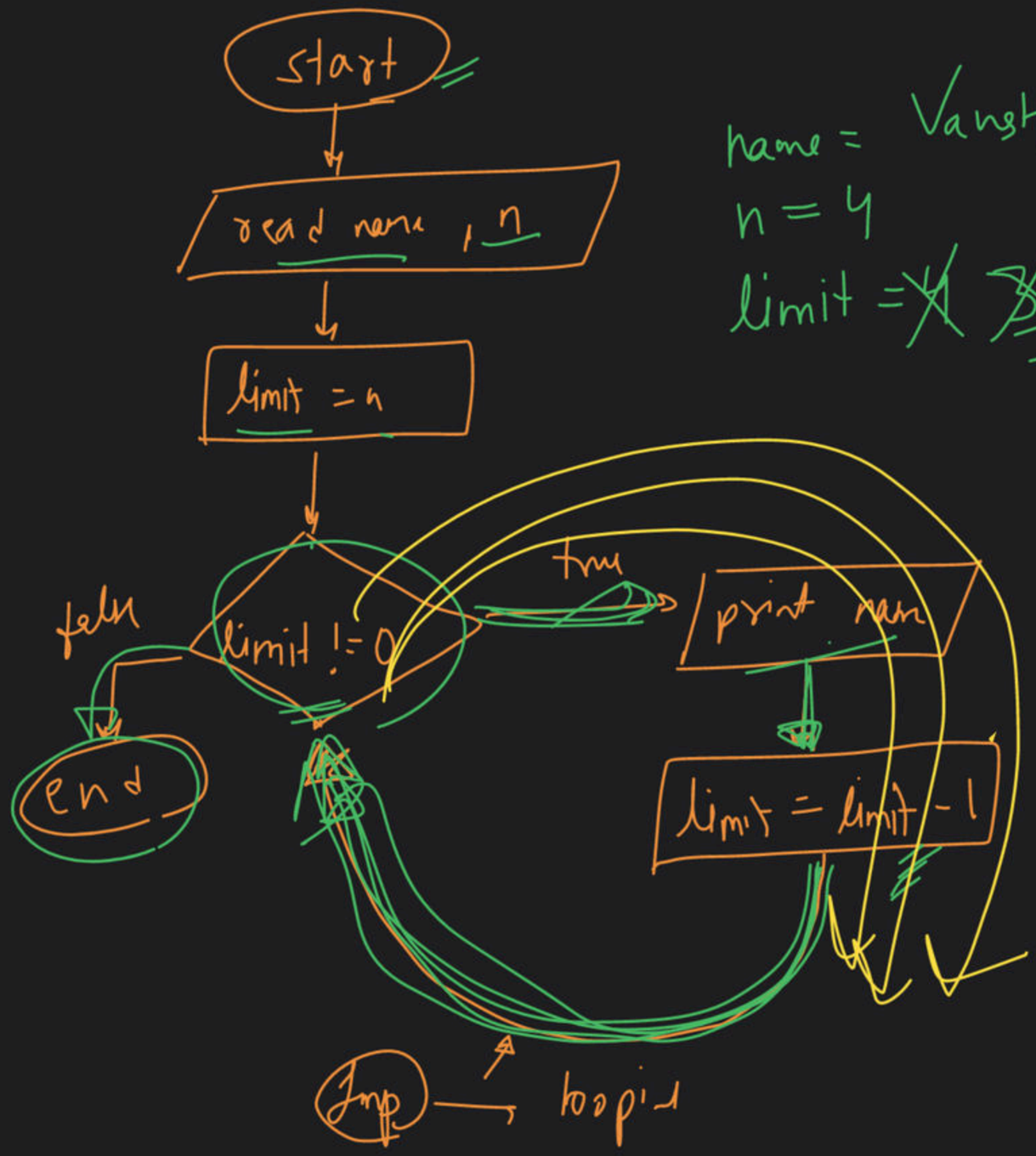

1 Start
2 read name
3 limit = 5
4 count = 0
5 if count < limit
print name
count = count + 1
else go to step 7
6 go to step 5
7 end

name = Babbar

limit :- 5 4 3 2 1 0

Babbar
Babbar
Babbar
Babbar
Babbar

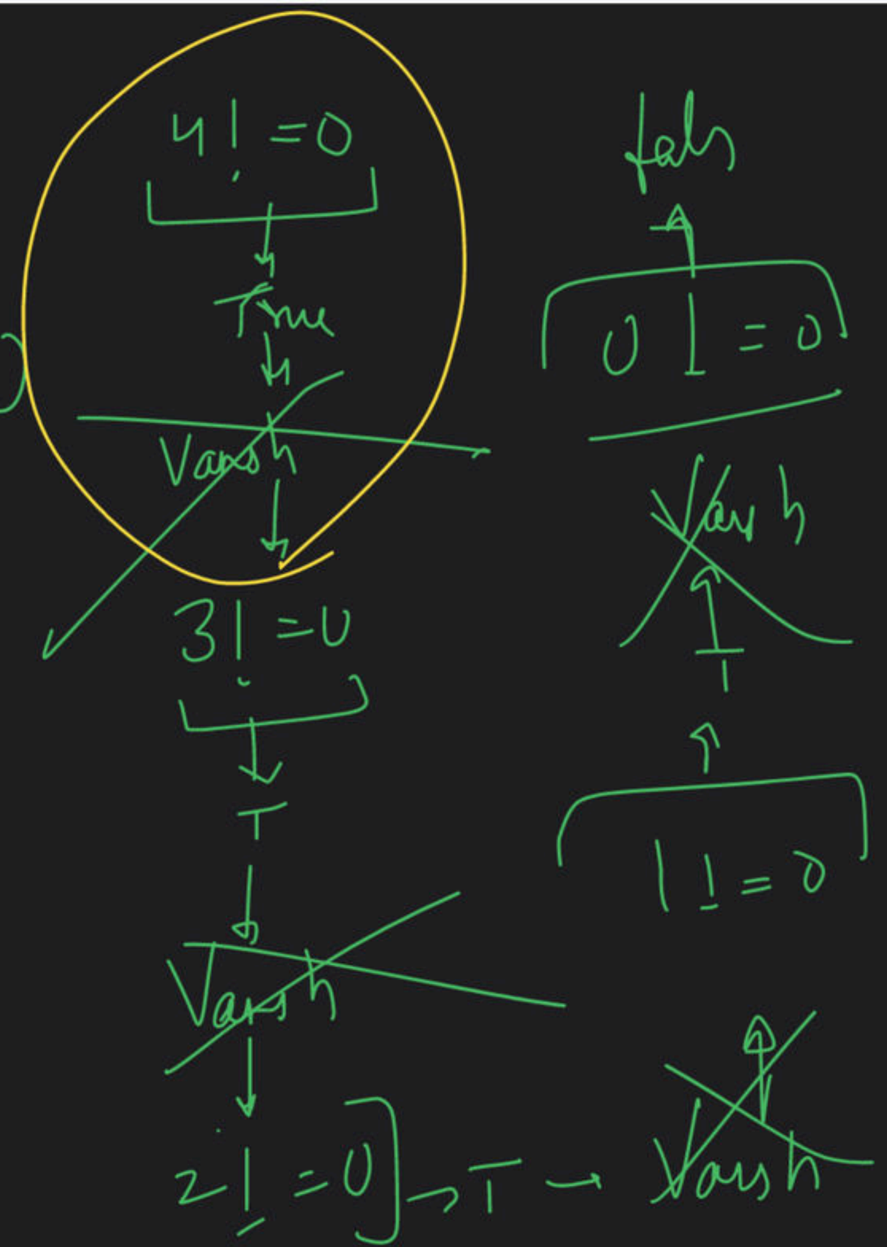




name = Vansh

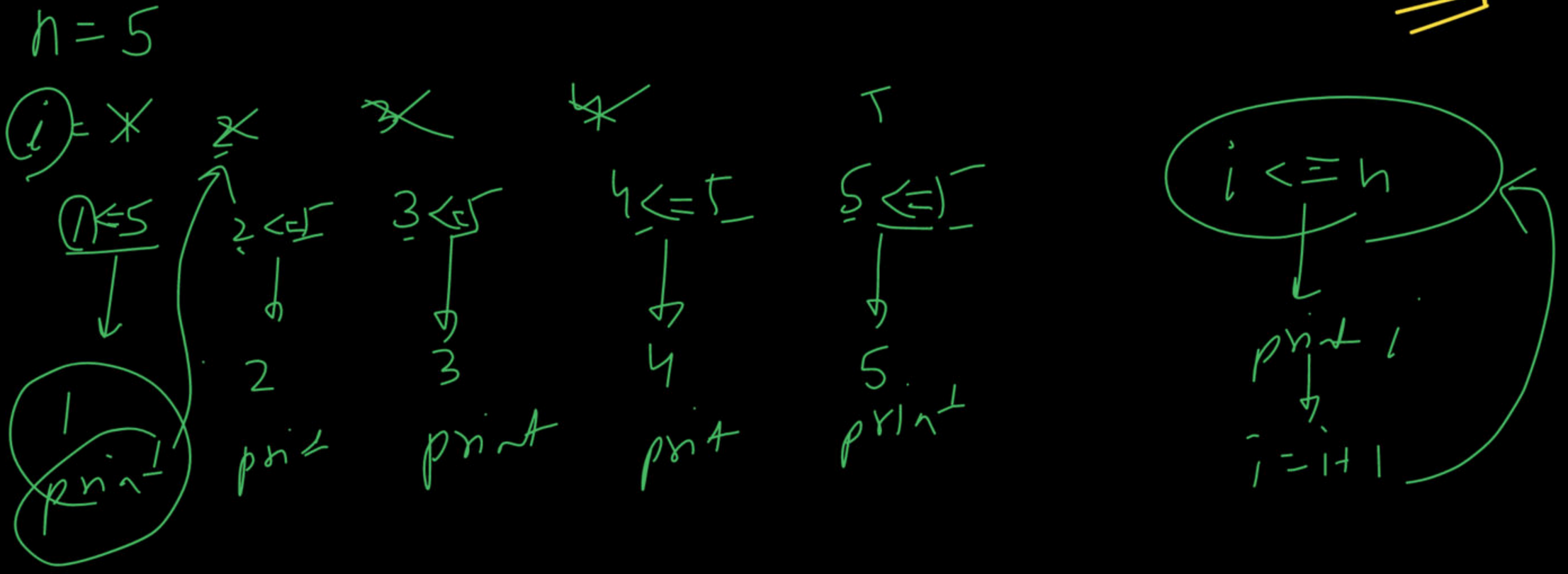
n = 4

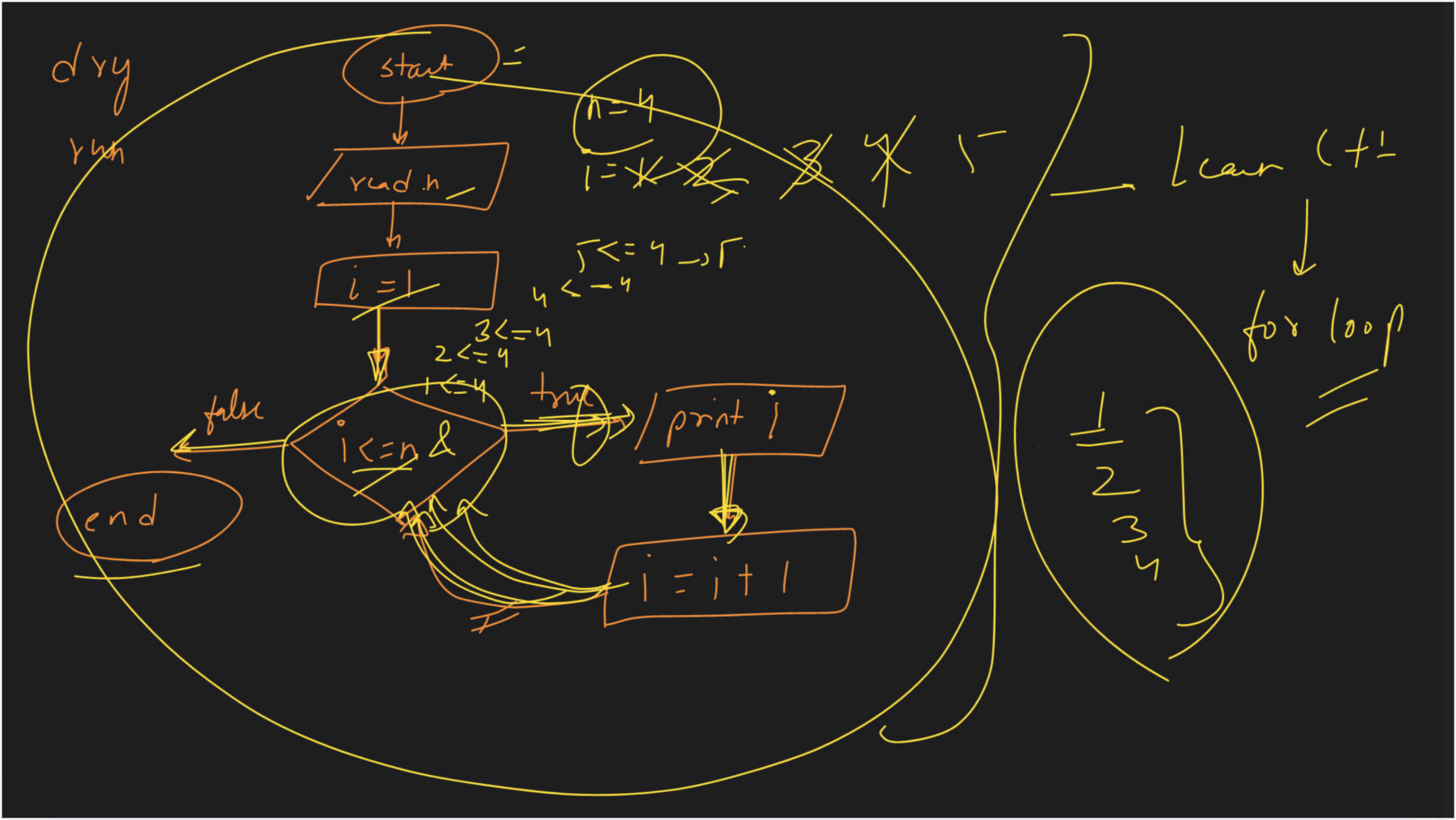
limit = ~~4~~ ~~3~~ ~~2~~ ~~1~~ 0

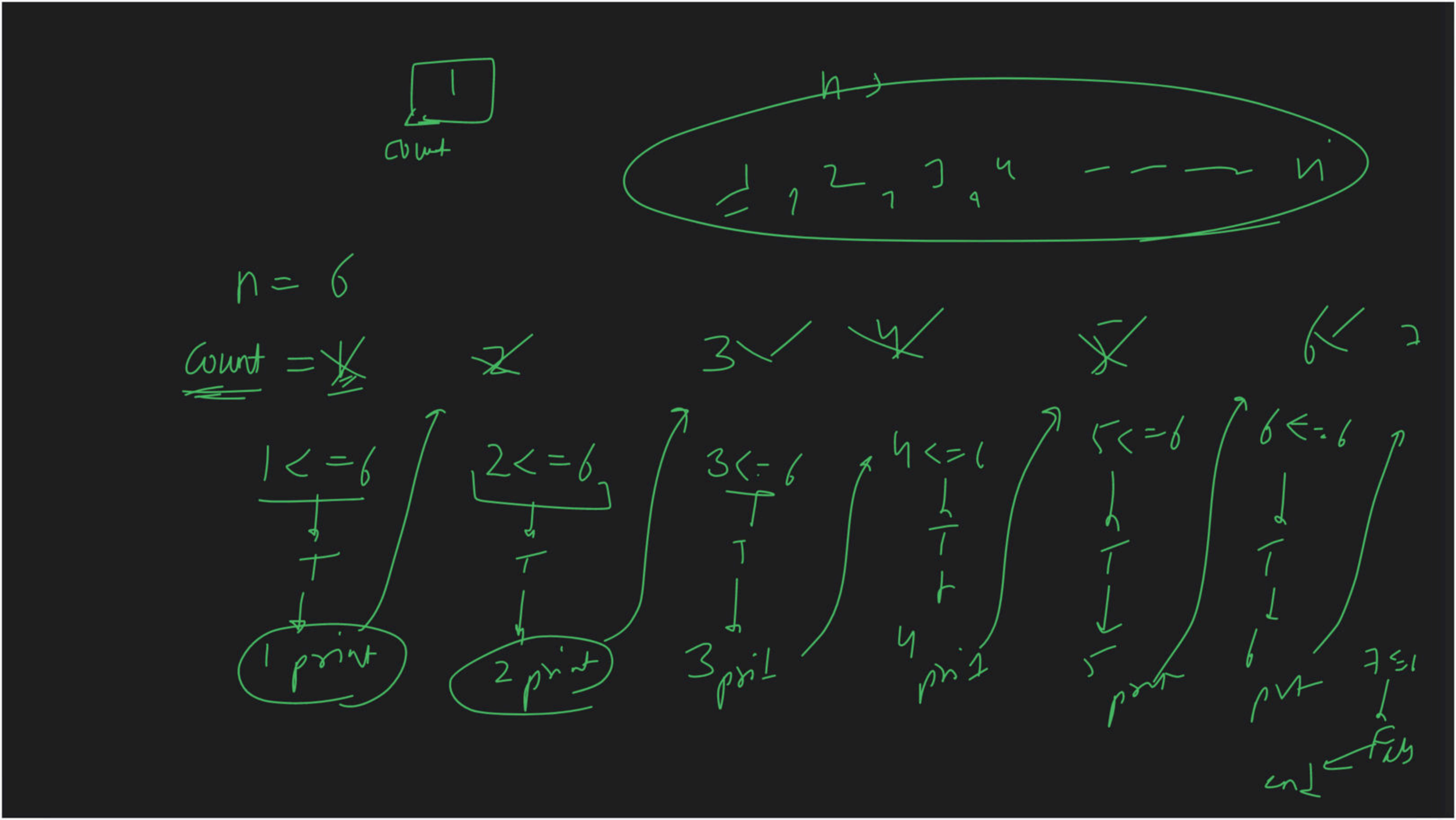


Design Flowchart - Print counting from 1 to N

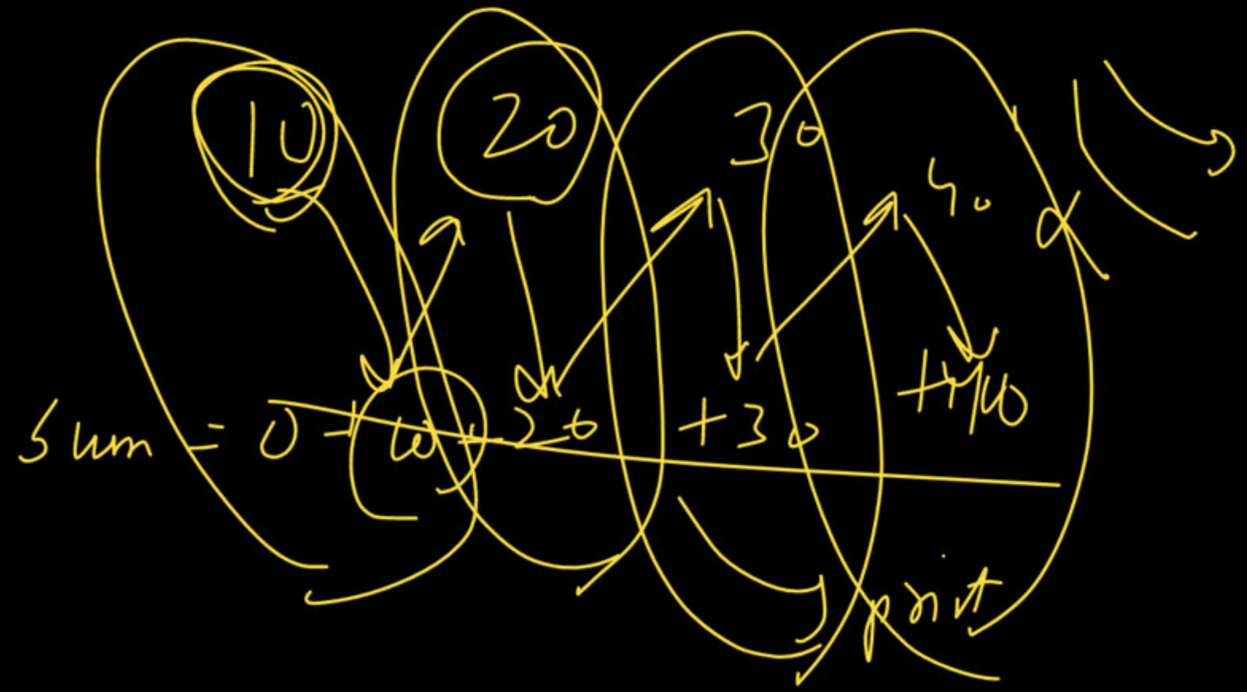
Loops

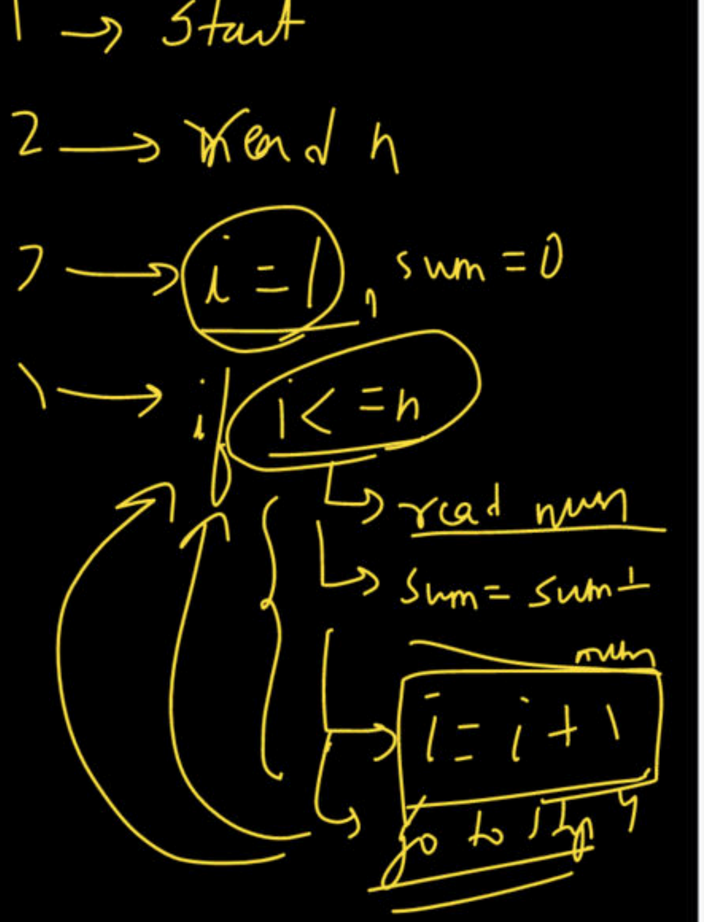


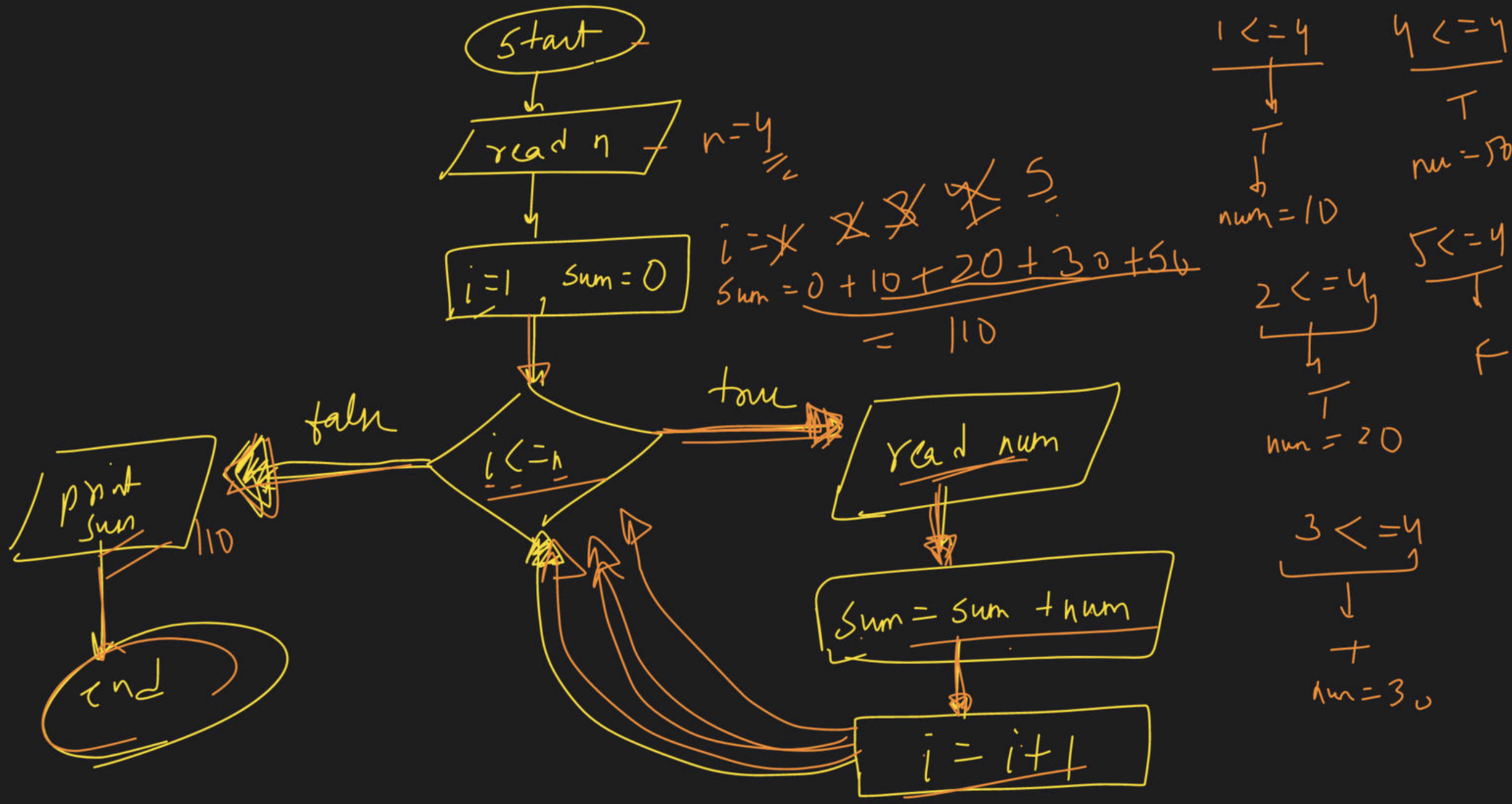

start
read n
i = 1
i <= n &
true
false
print i
i = i + 1
end
n = 4
for loop


1
count
n
1, 2, 3, 4 ---- n
n = 6
count = 1 2 3 4 5 6 7
1<=6
T
1 print
2<=6
T
2 print
3<=6
T
3 print
4<=6
T
4 print
5<=6
T
print
6<=6
T
print
7<=6
F
end

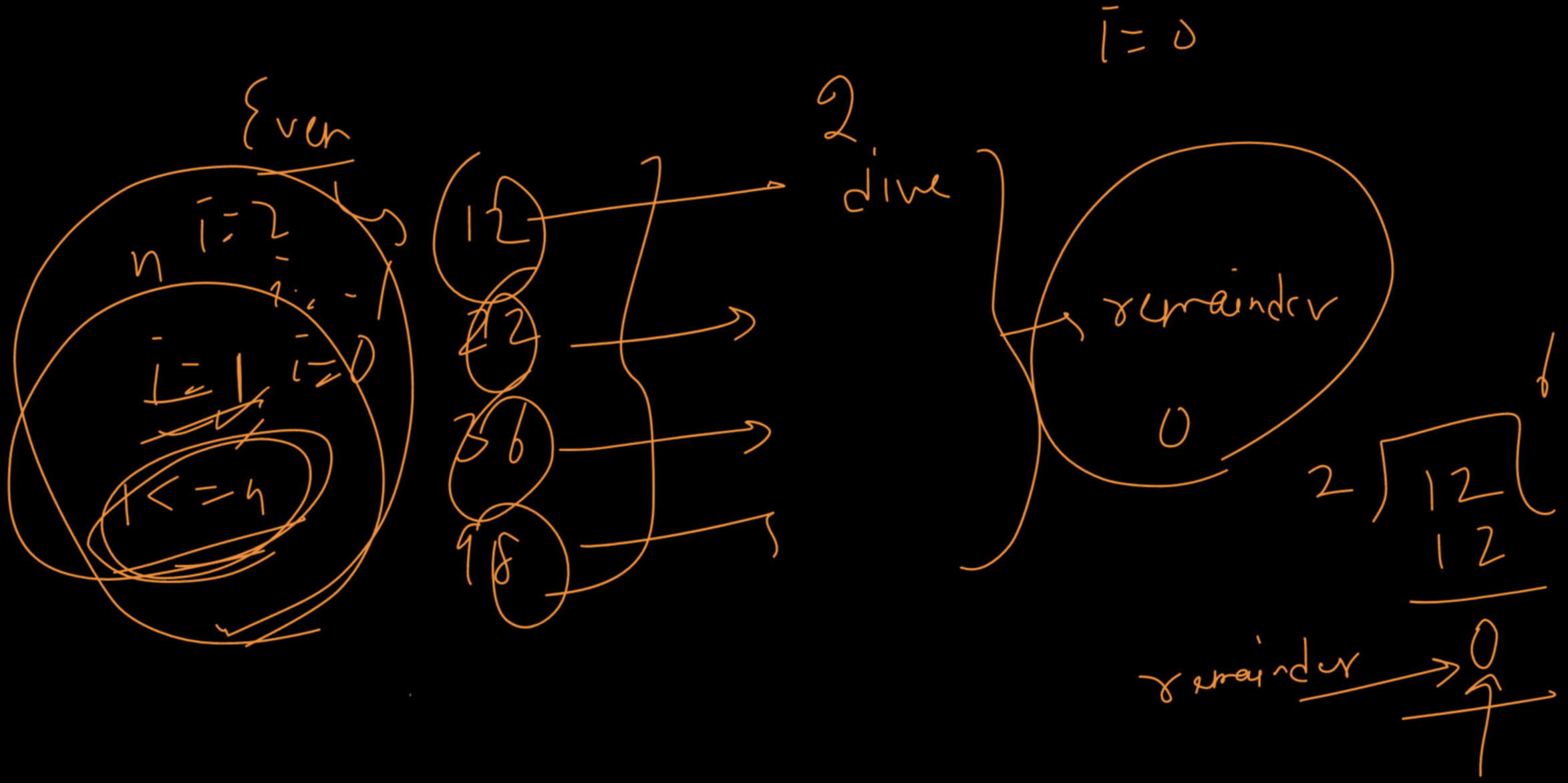
Design Flowchart - Add N numbers from user

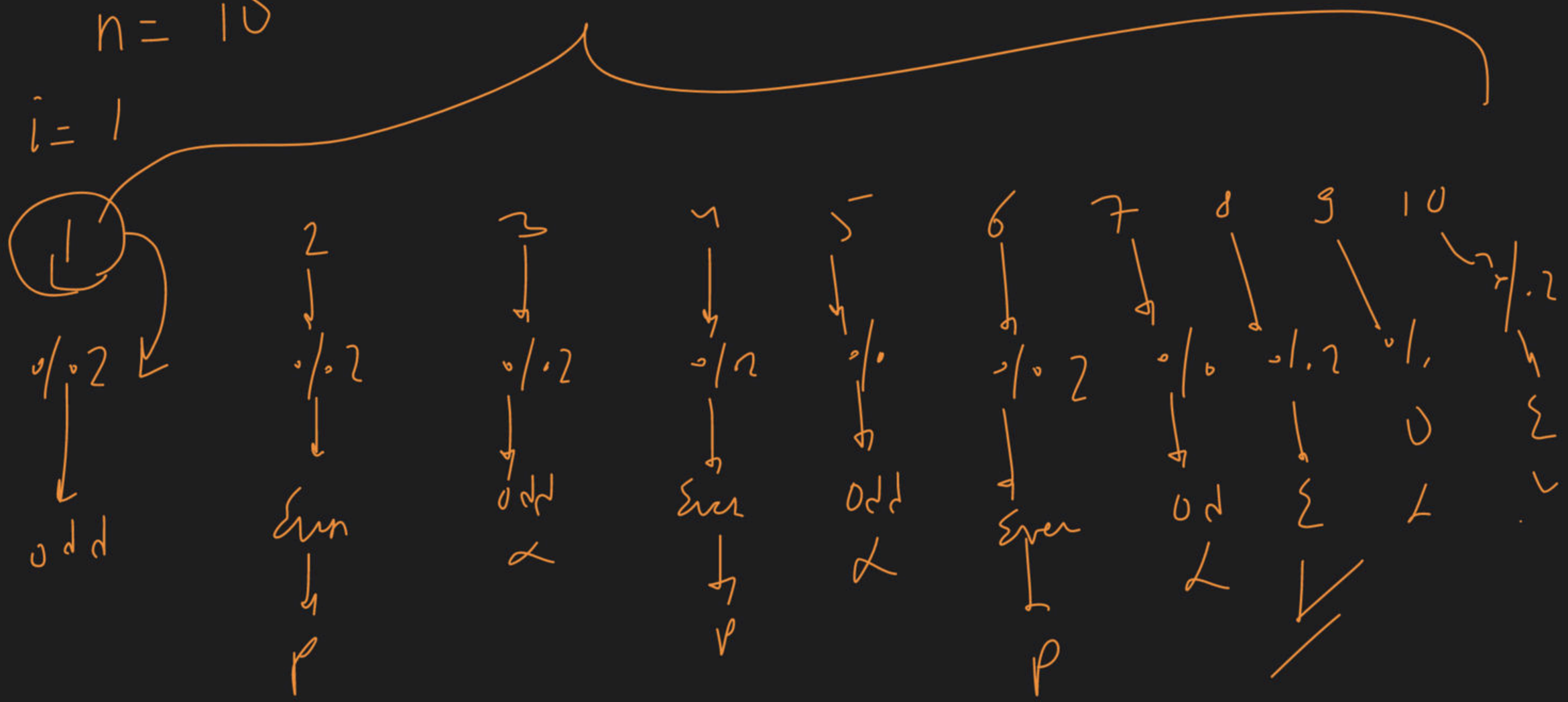


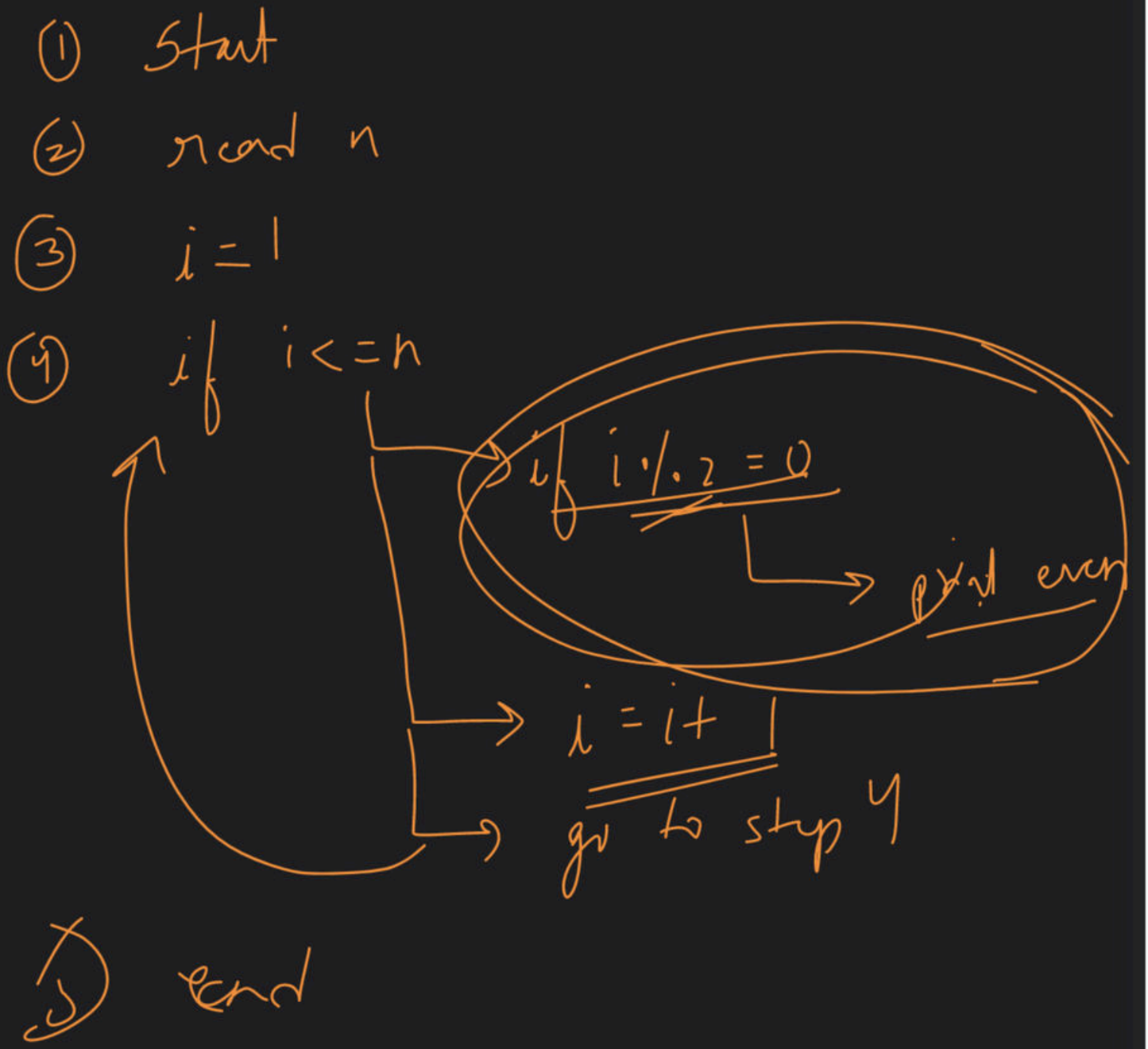


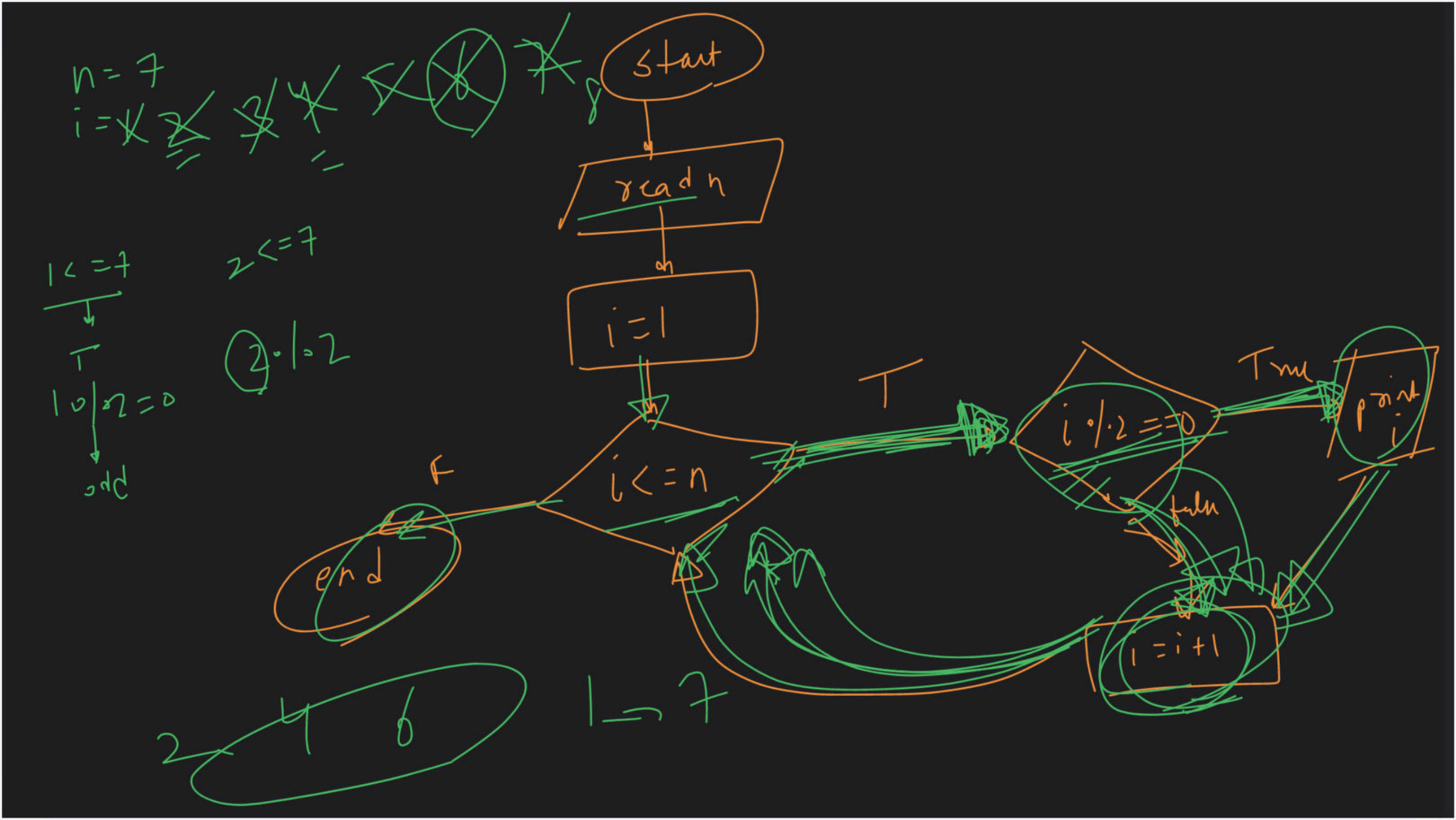

start
read n
n=4
i=1 sum=0
i<=n
false
true
print sum
110
end
read num
Sum = sum + num
i = i+1
i=X X X X 5
Sum = 0+10+20+30+50
= 110
1<=4
T
num=10
2<=4
T
num=20
3<=4
T
num=30
4<=4
T
5<=4
T
F

Design Flowchart - Print Even numbers from 1 to N




n = 10
i = 1
1
1
i < n
odd
2
i < n
Even
3
i < n
odd
4
i < n
Even
5
i < n
odd
6
i < n
Even
7
i < n
odd
8
i < n
9
i < n
10
i < n


1 Start
2 read n
3 i = 1
4 if i<=n
if i%2 = 0
print even
i = i+1
go to step 4
5 end


n=7
i=
1<=7
T
10%2==0
odd
2<=7
2.1.2
start
read n
i=1
i<=n
F
end
T
i%2==0
True
print i
false
i=i+1
2 4 6
1 => 7



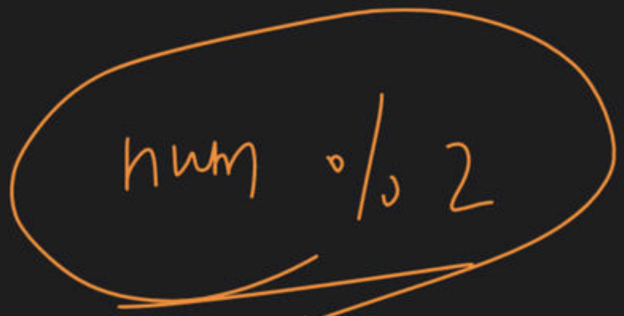
Modulo

Modulus

remainder

$12 \% 2 = 0$

rem



→ 0 → Even

→ !0 → Odd

Assignment:

Design Flowchart for below:

- • Multiply 2 number after taking input
- • Perimeter of triangle
- • Find Simple Interest
- • Find Compound Interest
- • Print Counting from N to 1
- • Find Factorial of a Number
- • Check if a number is Prime or not
- • Check Valid Triangle or not
- • Print Max of 2 number

